

Computer Vision 1

Object Recognition

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UNIVERSITY
OF AMSTERDAM



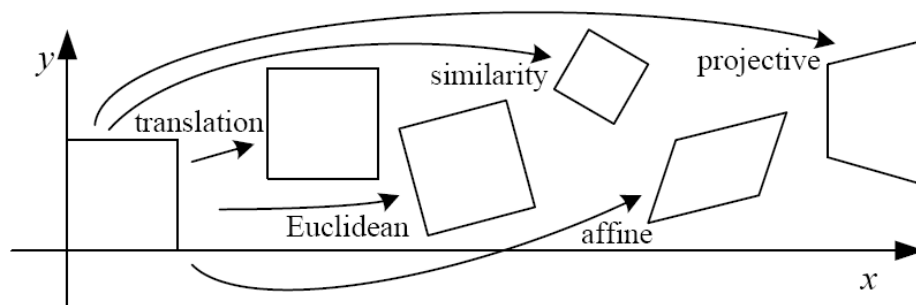
computer
vision lab

Lecture overview

Week		Date	Presenter		Lectures	Excerise / Labs
1	36	02/09/2024	Arnoud Visser		1. Introduction and Course Logistics; Camera model	
1	36	05/09/2024	Arnoud Visser		2. Sensor and image formation Color	
2	37	09/09/2024	Arnoud Visser		3. In place image processing Photometric stereo	
2	37	12/09/2024	Arnoud Visser		4. Convolution and image filtering	
3	38	16/09/2024	Arnoud Visser		5. Feature: edge and corner	
3	38	19/09/2024	Arnoud Visser		6. Motion and optical flow	
4	39	23/09/2024	Martin Oswald		7. Global transform, Image stitching, alignment	
4	39	26/09/2024	Martin Oswald	Today	8. Basics of Recognition 1	
5	40	30/09/2024	Martin Oswald	Online	9. Basics of Recognition 2	
5	40	03/10/2024	Martin Oswald	Lectures	10. Convolutional neural network 1 (Basic)	
6	41	07/10/2024	Martin Oswald		11. Convolutional neural network (Training and testing)	
6	41	10/10/2024	Martin Oswald		12. Stereo, 3D reconstruction	
7	42	14/10/2024	TBA		13. Guest lectures	
7	42	17/10/2024	TBA; Martin & Arnoud		14. Guest lecture; Course Wrap-up, Exam Q&A	
8	43	21/10/2024	Your turn!		Final exam	

Recap

Basic 2D Transformations



Name	Matrix	# D.O.F.	Preserves:	Icon
translation	$\begin{bmatrix} I & t \end{bmatrix}_{2 \times 3}$	2	orientation + ...	
rigid (Euclidean)	$\begin{bmatrix} R & t \end{bmatrix}_{2 \times 3}$	3	lengths + ...	
similarity	$\begin{bmatrix} sR & t \end{bmatrix}_{2 \times 3}$	4	angles + ...	
affine	$\begin{bmatrix} A \end{bmatrix}_{2 \times 3}$	6	parallelism + ...	
projective	$\begin{bmatrix} \tilde{H} \end{bmatrix}_{3 \times 3}$	8	straight lines	

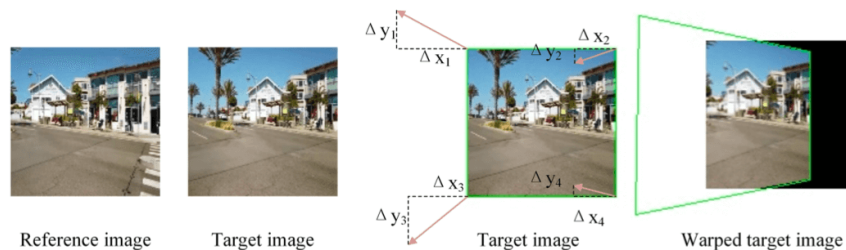
These transformations are a nested set of groups.
Closed under composition and inverse is a member.

Overview: Matrix Groups

Set of square $n \times n$ matrices	General Linear Group	Orthogonal Group	Special Orthogonal Group
	$\mathcal{M}(n) \supset GL(n) = \{A \in \mathcal{M}(n) \mid \det(A) \neq 0\}$ Invertible square matrices	$O(n) = \{R \in GL(n) \mid R^T R = I\}$ Orthogonal matrices	$SO(n) = \{R \in O(n) \mid \det(R) = 1\}$ Rotation matrices

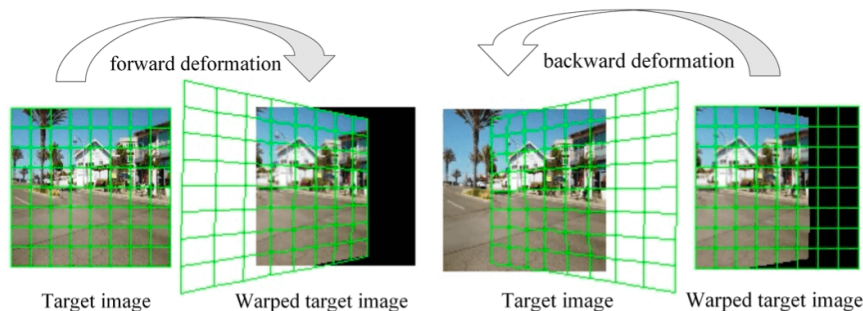
General Linear Group	Affine Group	Euclidean Group	Special Euclidean Group
	$GL(n+1) \supset \mathcal{A}(n) = \left\{ \begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \mid A \in GL(n), b \in \mathbb{R}^n \right\}$ Affine matrices	$E(n) = \left\{ \begin{pmatrix} R & t \\ 0 & 1 \end{pmatrix} \mid R \in O(n), t \in \mathbb{R}^n \right\}$	$SE(n) = \left\{ \begin{pmatrix} R & t \\ 0 & 1 \end{pmatrix} \mid R \in SO(n), t \in \mathbb{R}^n \right\}$ Rigid-body motions

Forward vs. Backward Warping



(a) Input images.

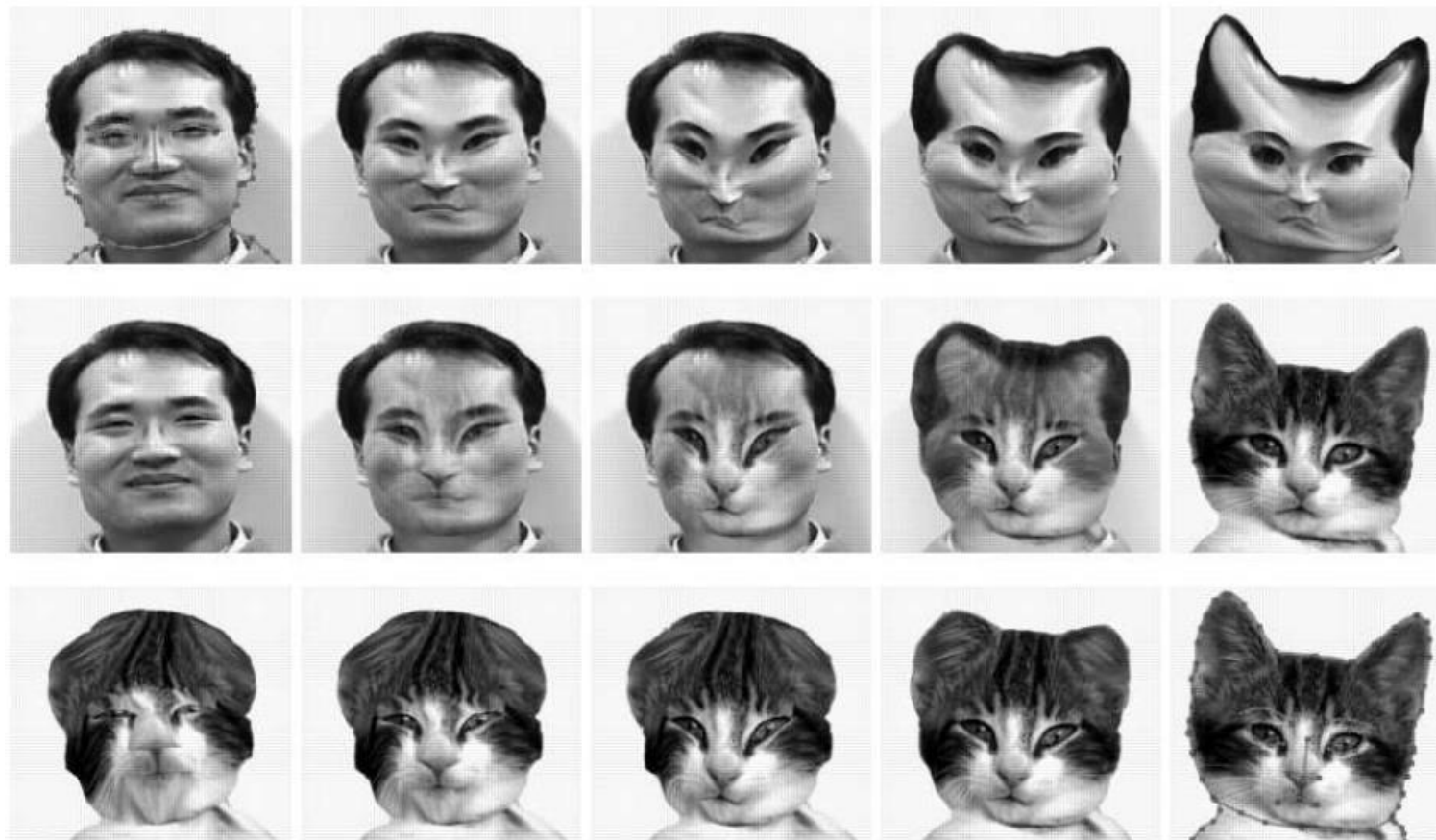
(b) Forward single grid deformation.



(c) Forward multi-grid deformation. (d) Backward multi-grid deformation.

Fig. 6. Comparison between the forward deformation and backward deformation.

Image Morphing



warp

man to cat

blend

morphing

blend

warp

cat to man

Image Stitching

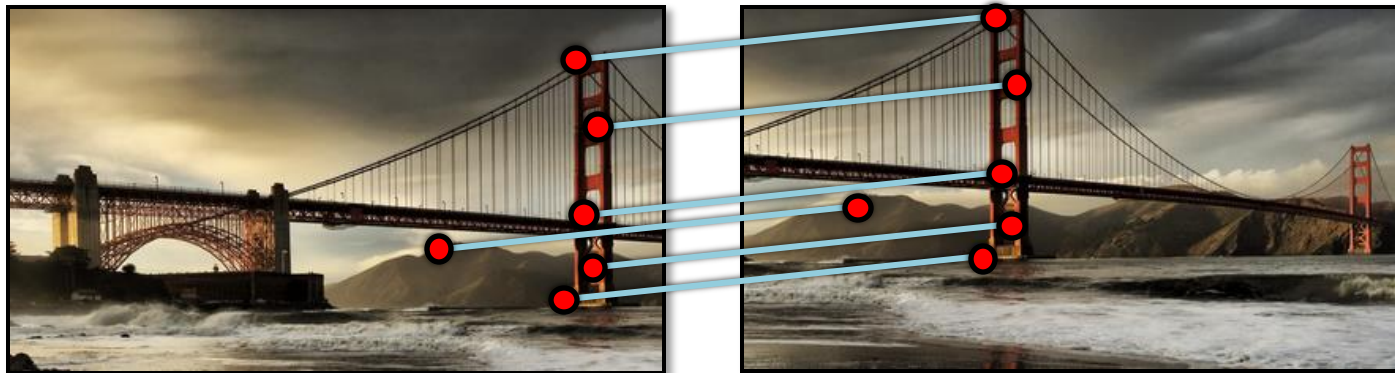


Goal: Combine two or more overlapping images into one large image.

Typical Steps:

- Transformation computation
 - Reprojection model
 - Key point matching
 - RANSAC
 - Solving transformation
- Planar mapping
- Blending

Simple Case: Translation



(x_t, y_t)

How do we solve for
 (x_t, y_t) ?

Solving for Translations

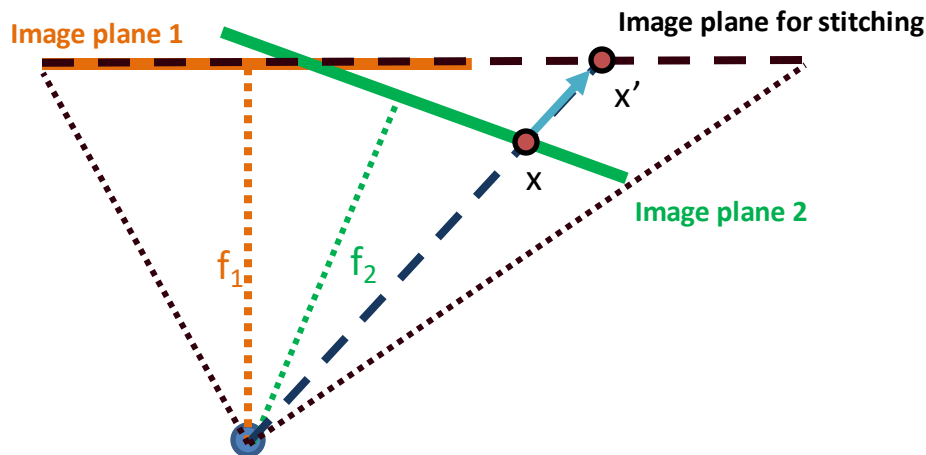
$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ \vdots & \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_t \\ y_t \end{bmatrix} = \begin{bmatrix} x'_1 - x_1 \\ y'_1 - y_1 \\ x'_2 - x_2 \\ y'_2 - y_2 \\ \vdots \\ x'_n - x_n \\ y'_n - y_n \end{bmatrix}$$

$$\underset{2n \times 2}{\mathbf{A}} \underset{2 \times 1}{\mathbf{t}} = \underset{2n \times 1}{\mathbf{b}}$$

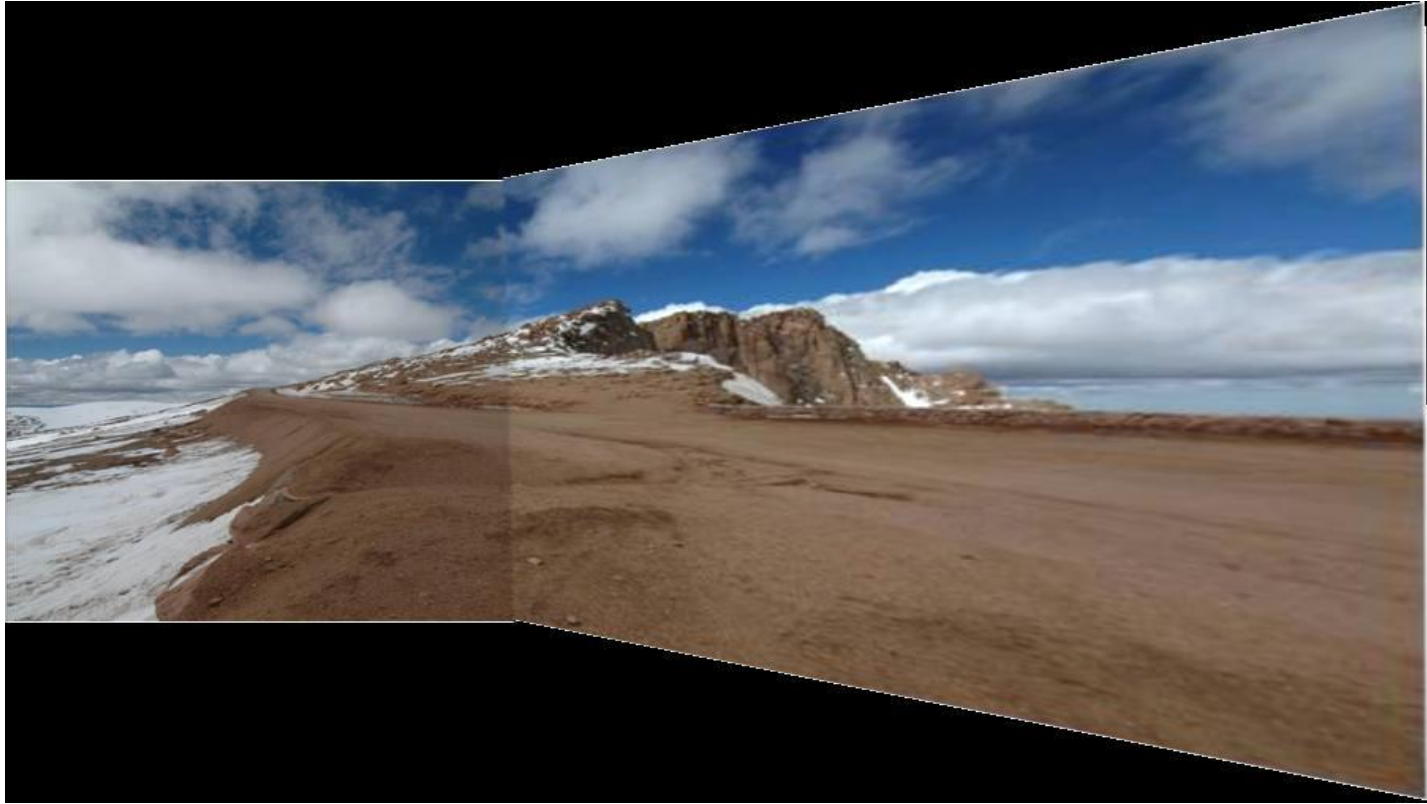
→ Model fitting by solving a linear system (after removing potential outliers with RANSAC).

Planar Projection (Homography)

- **Goal:** Reproject several images into a new stitched image.
Here, the image domain is **planar**, but could also be cylindrical, spherical, etc.
- **Approach:** **projective mapping** (homography)
describes how coordinates from one image plane transform to coordinates in another image plane.



Planar Projection (Homography)



[Photo by Russ Hewett]

Blending Comparison



(b) Without gain compensation



(c) With gain compensation



(d) With gain compensation and multi-band blending

- Chapter 6 (until 6.2.2)

Image Understanding

AN IMAGE IS WORTH A THOUSAND WORDS



Image Understanding

Classification



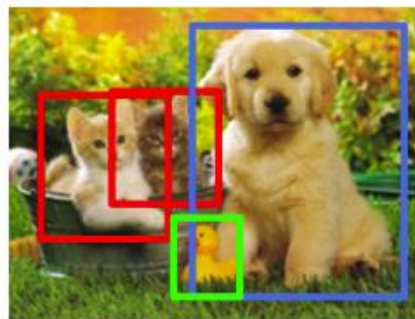
CAT

**Classification
+ Localization**



CAT

Object Detection



CAT, DOG, DUCK

**Instance
Segmentation**



CAT, DOG, DUCK

- Object Classification: classify the main object in an image if the image has one main object.
- Visual Image Retrieval: retrieve images from the library through a query image.
- Object Detection: identify whether there is an object in an image and recognize the object.
- Image Segmentation: segment an image into regions according to the pixels the object belongs to.

Today: Object Classification!

Today's Agenda



- **Object Classification**
 - Feature Representation
 - Bag of words
 - Feature space & Decision boundary
 - Classification Performance Evaluation
 - Multi-class Classification
 - Datasets

Computer Vision is hard!

Challenges: Similar Appearance



Challenges: Similar Appearance



Challenges: Similar Appearance



Challenges: Similar Appearance



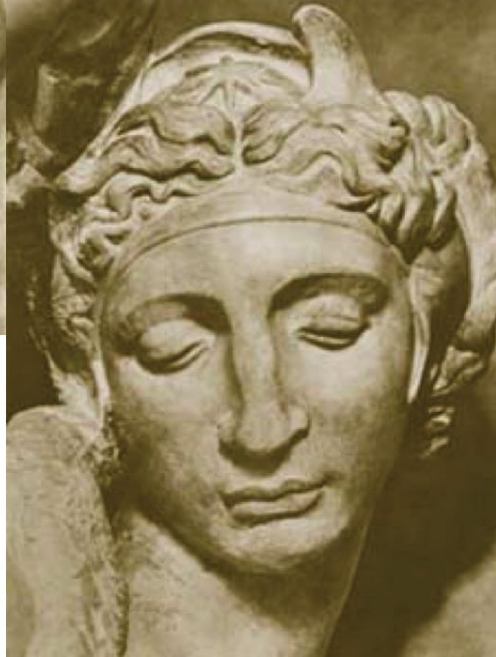
Challenges: Similar Appearance



Challenges: Similar Appearance



Challenges: View Point Variation



Michelangelo 1475-1564

slide by Fei Fei, Fergus & Torralba

Challenges: Illumination Changes



Challenges: Occlusion



Magritte, 1957

slide by Fei Fei, Fergus & Torralba

Challenges: Scale Changes



slide by Fei Fei, Fergus & Torralba

Challenges: Intra-class Variations



Challenges: Number of Variables



$1920 \times 1080 \times 3 = 6,220,800$ input variables

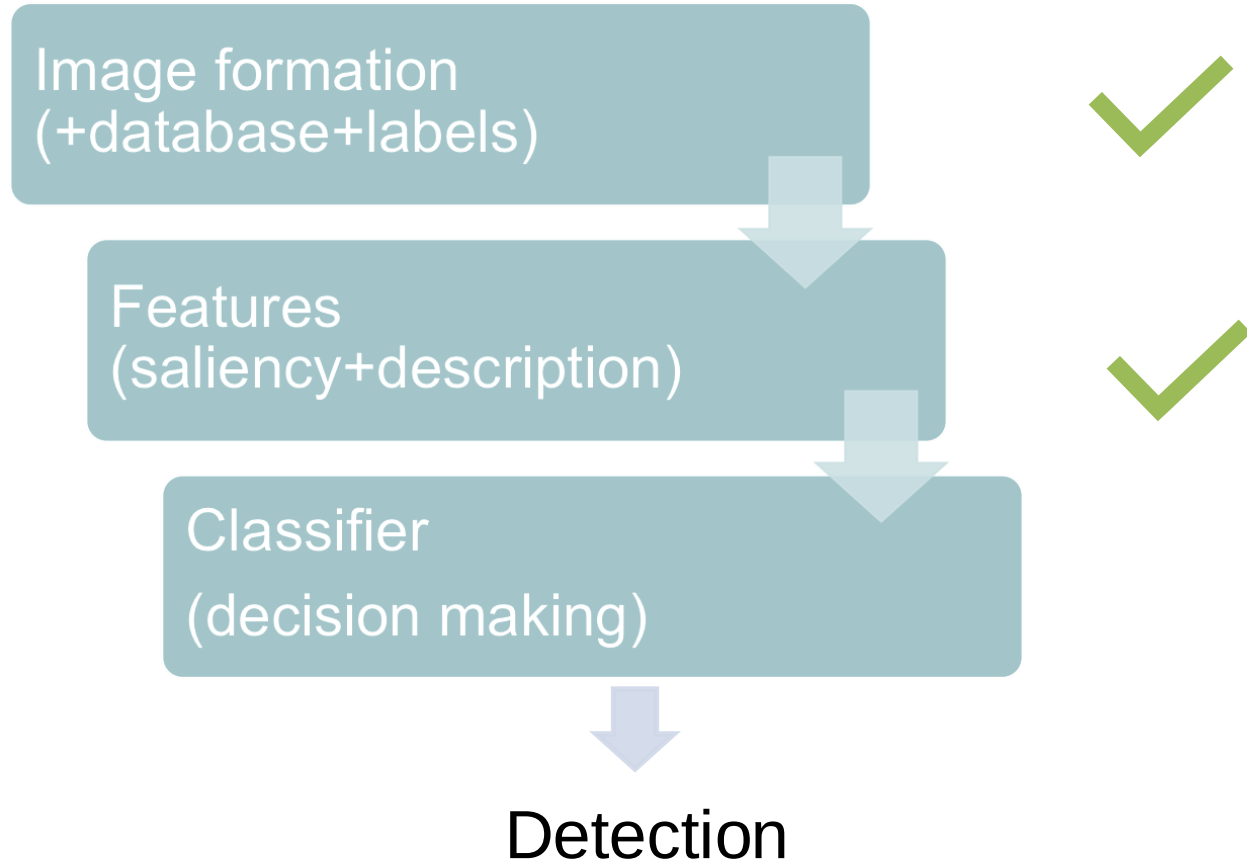
Computer Vision ~~is~~ hard!
used to be ?

What makes image classification difficult?



- Huge dimensionality
 - A 256x256 RGB image amounts to ~200K input variables
- Image formation process
 - 2D summary of 3D world
 - Variances in lighting, orientations, scale
 - Scene layout, cluttering, occlusions

Classification Steps



Today's Agenda

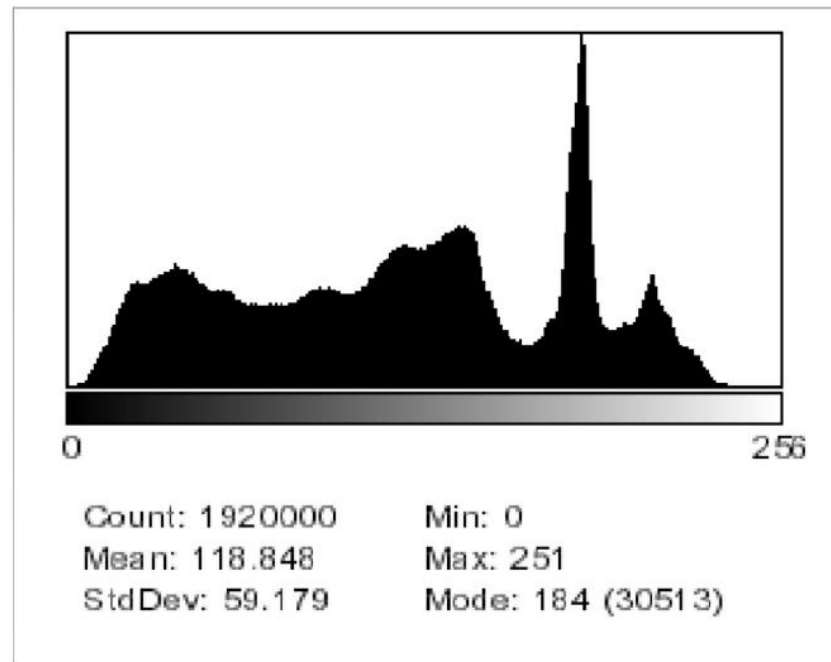


- **Image Classification**
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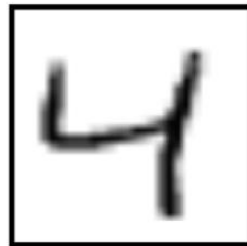
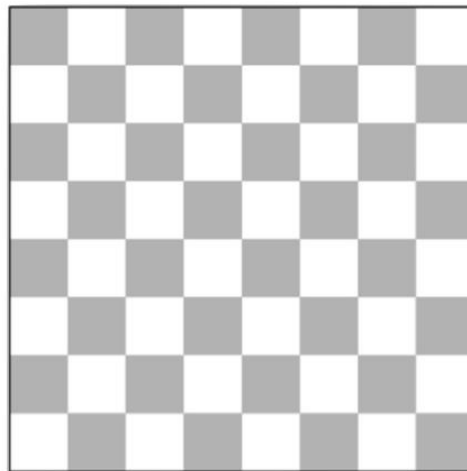
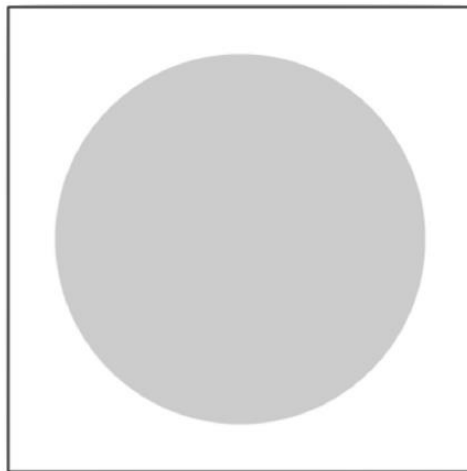
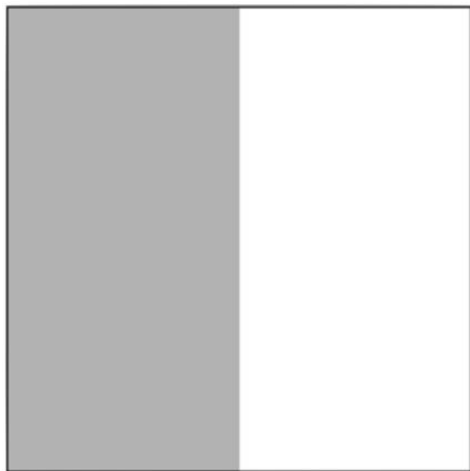
Image and Feature Representations

- Pixel representations
- Global representations
- Local features
 - Harris Corners / SIFT features
 - HoG feature descriptor

Global Histogram



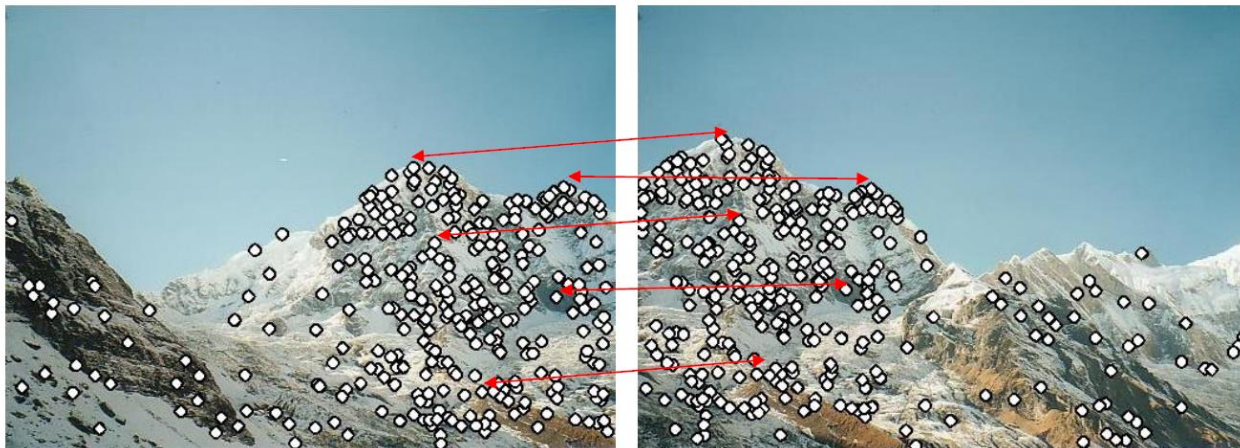
Global Histogram





Is this the same mountain?

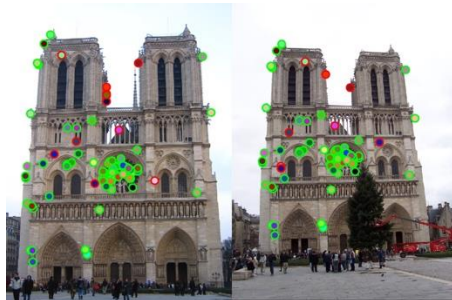
Local Features



Feature Points: Detectors vs. Descriptors

Feature Detector

(Finding distinct points)

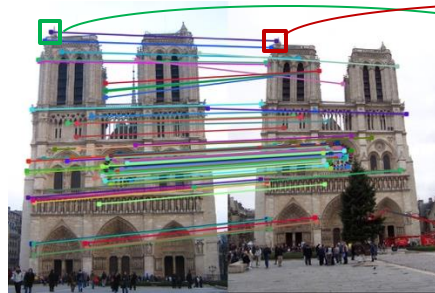


[S. Shah, [Local Feature Matching](#)]

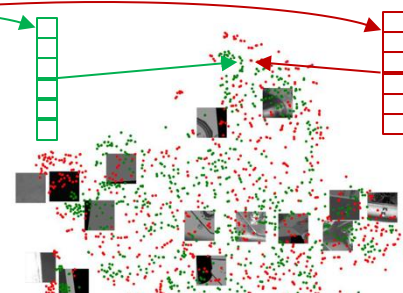
- **Harris** corners
- **Shi-Tomasi** corner detector
- **SUSAN** corner detector
- **FAST** feature detector
- **Laplacian of Gaussian**
- **Difference of Gaussian**
- **Superpoint**
- ...

Feature Descriptor

(Describing a keypoint)



[S. Shah, [Local Feature Matching](#)]



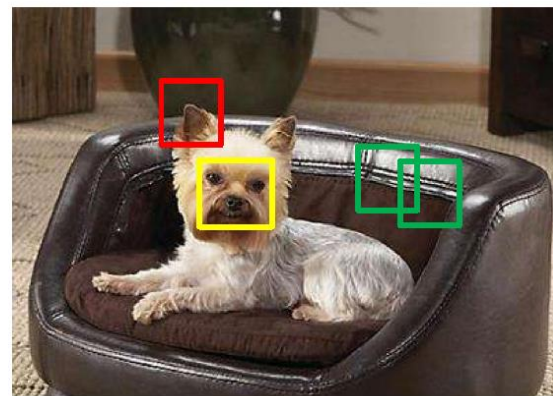
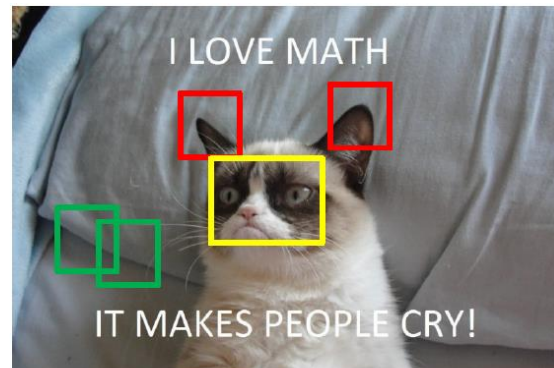
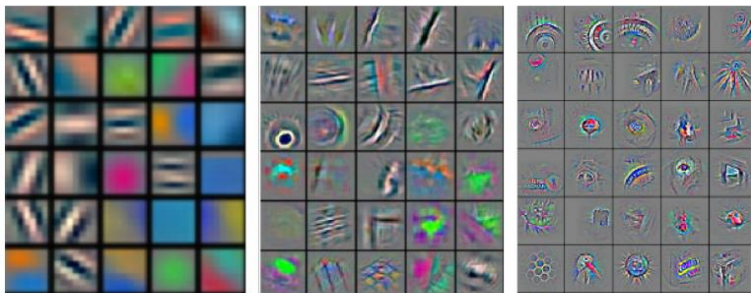
[Ghimire et al., Applied Sciences, 2021]

- **SIFT** – Scale Invariant Feature Transform
- **SURF** – Speeded-up Robust Features
- **BRIEF** – Binary Robust Independent Elementary Features
- **ORB** – Oriented FAST Rotated BRIEF
- **Superpoint**
- ...

Why Detect Local Features?

If we sample enough images

... we would see very similar filters no matter
their location



Local image features reoccur

Why Detect Local Features?

- Desired Properties:
 - Scale Invariance
 - Rotation Invariance
 - Robust to Lighting Changes
 - Robust Projective Changes
 - Etc.

Image Representations - Summary

- Pixel values: no translation
- Global histograms:
 - weak discriminative power
- Local features:
 - Should be used to recognize objects in images
 - How to create a representation from these local features

Today's Agenda

- **Image Classification**
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 - **Bag of words**
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How many features are needed?

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!

Text case

How many features are needed?

The Bag of Words Representation

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!

15



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
...	...

How many features are needed?

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!

Text case

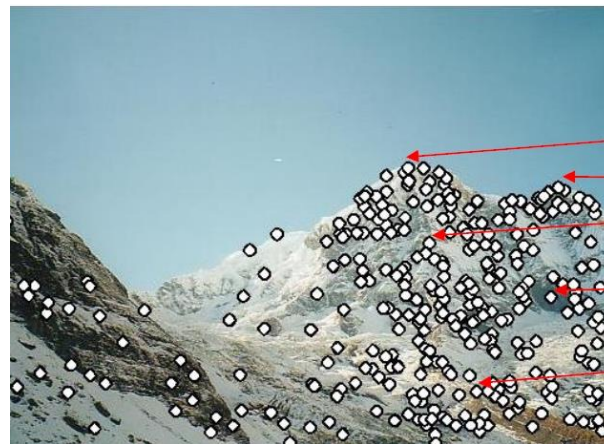
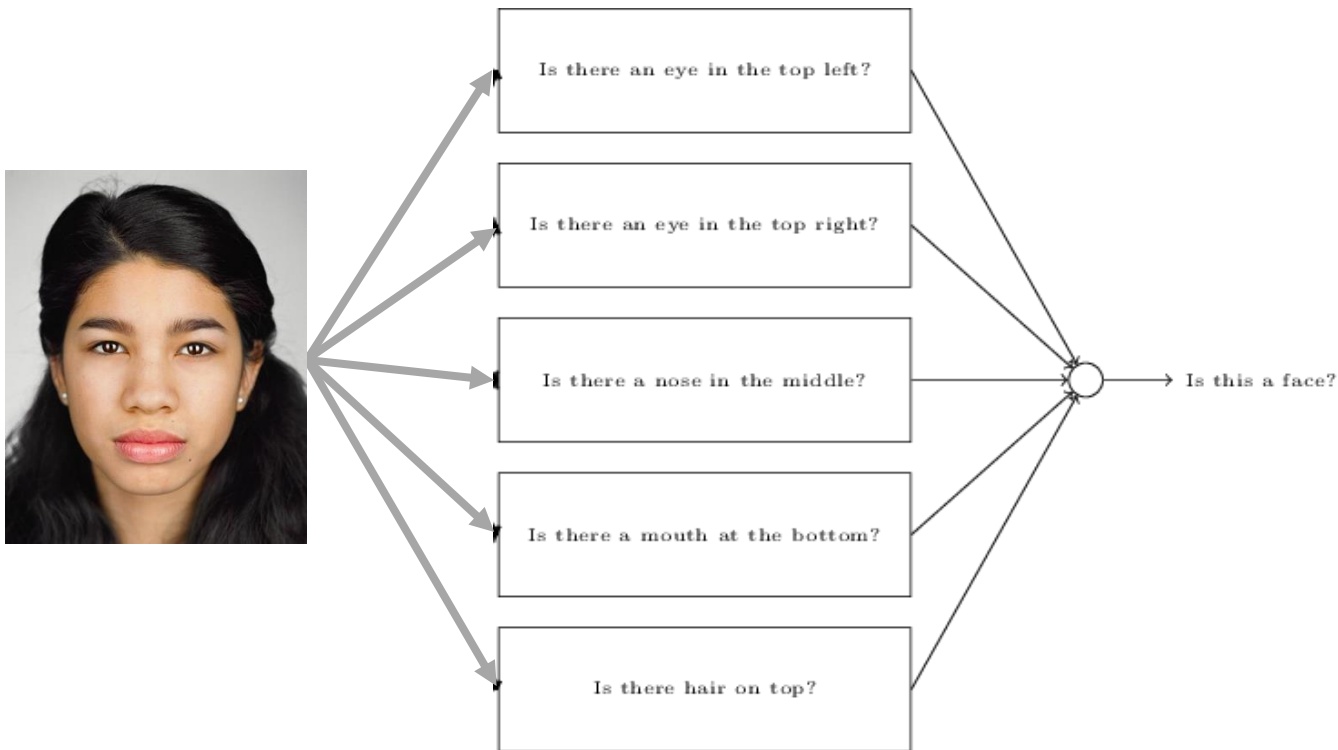
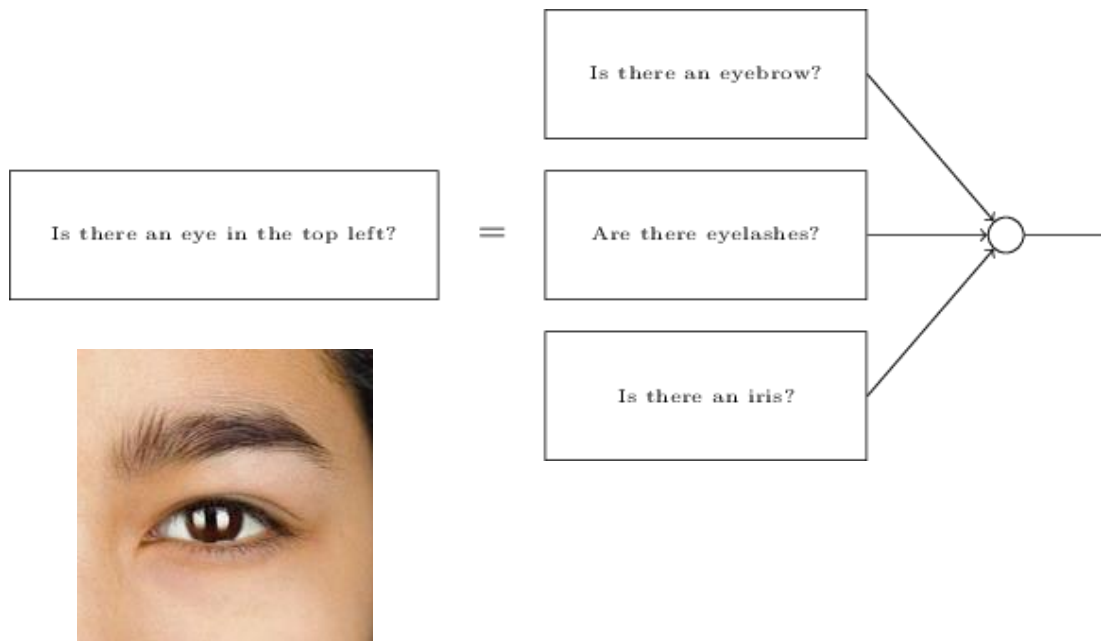


Image case

Example



Example

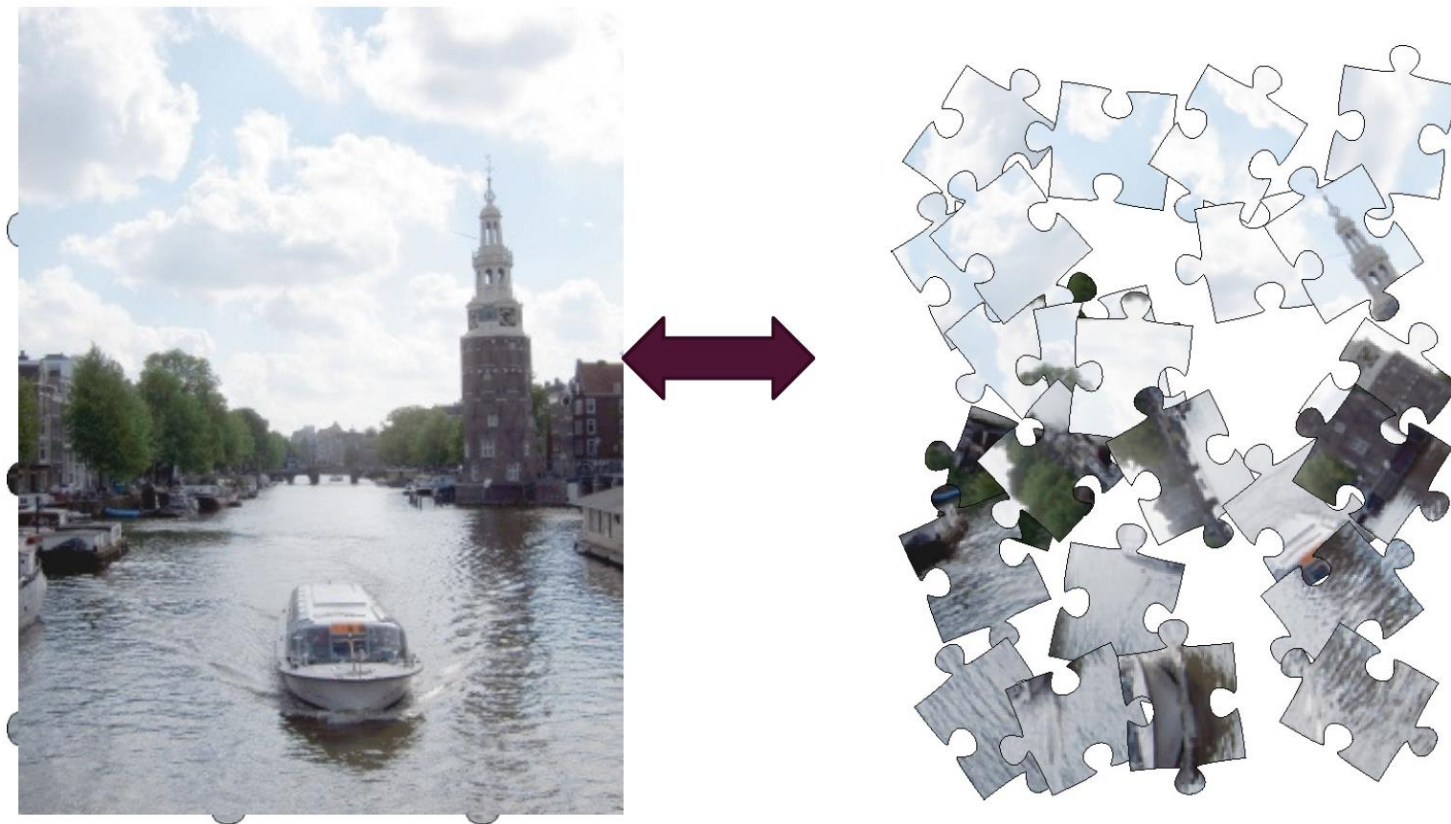


Bag of Visual Words

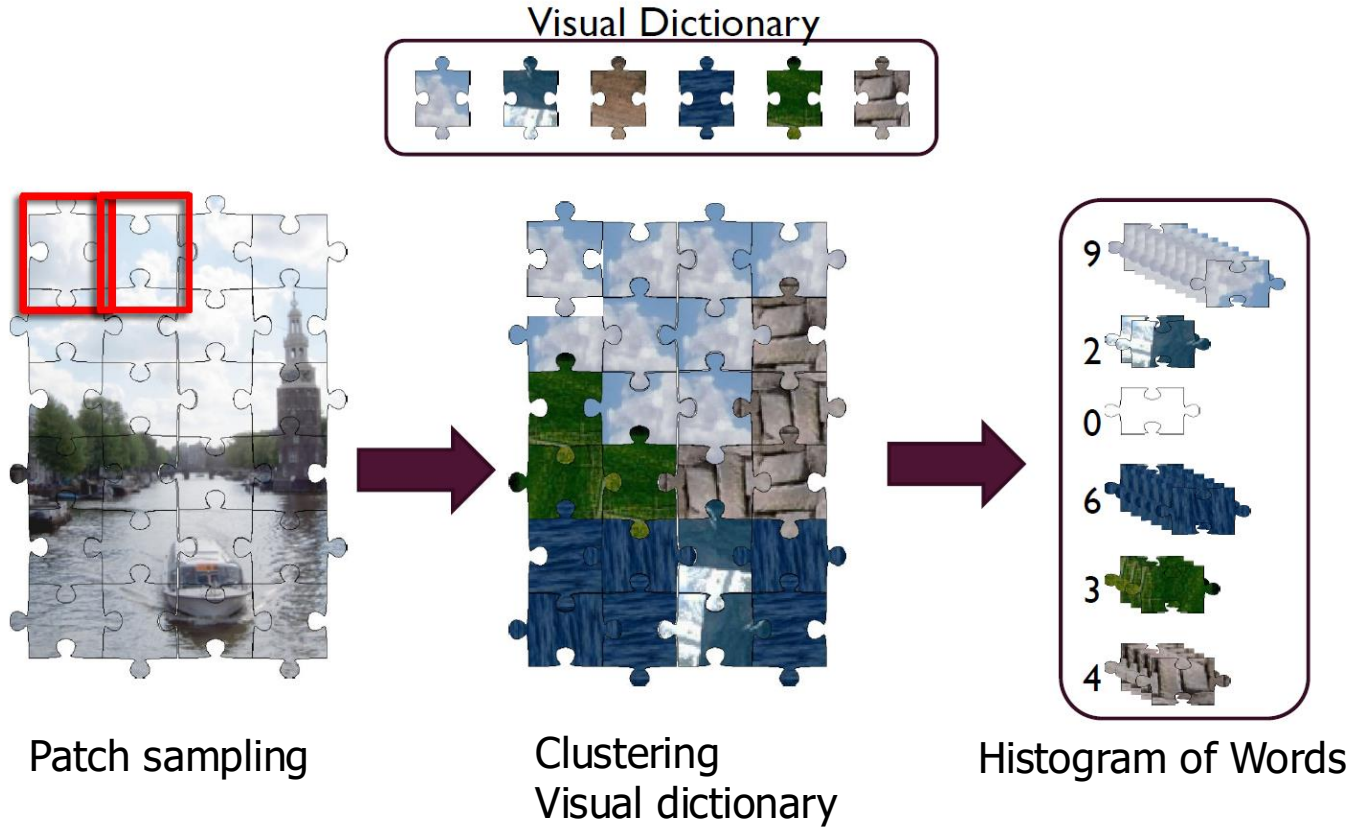


Slide credits: Jan van Gemert

Images as Bag of Patches



Images as Bag of Patches

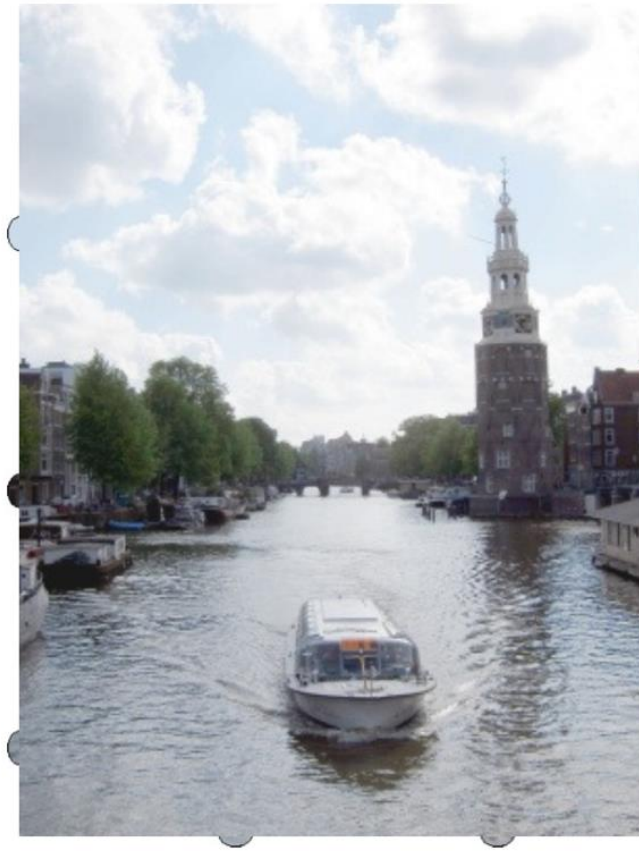


Bag of Words: Key Ingredients

1. Patch sampling
2. Patch descriptors
3. Visual Dictionary
4. Histogram creation

There are choices to be made!

Patch Sampling



Patch Sampling

SIFT

RGB

HOG

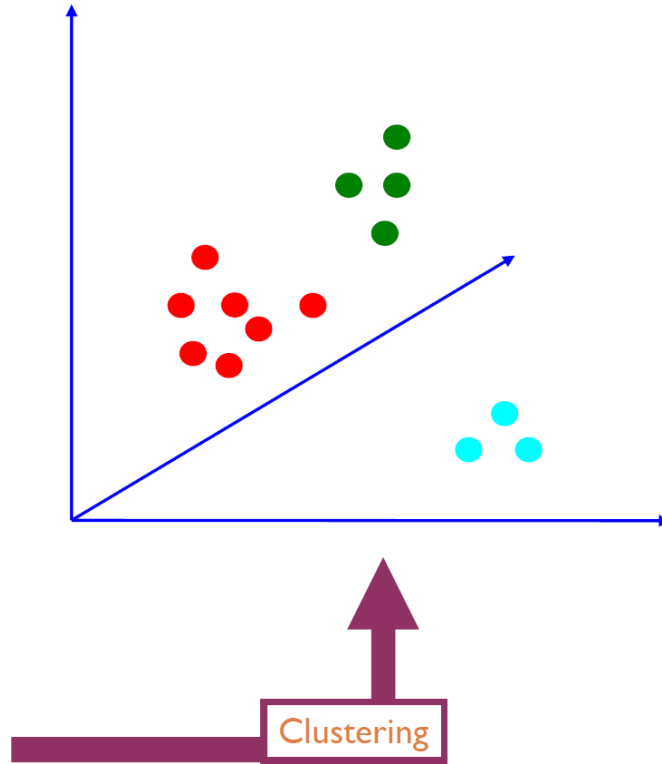
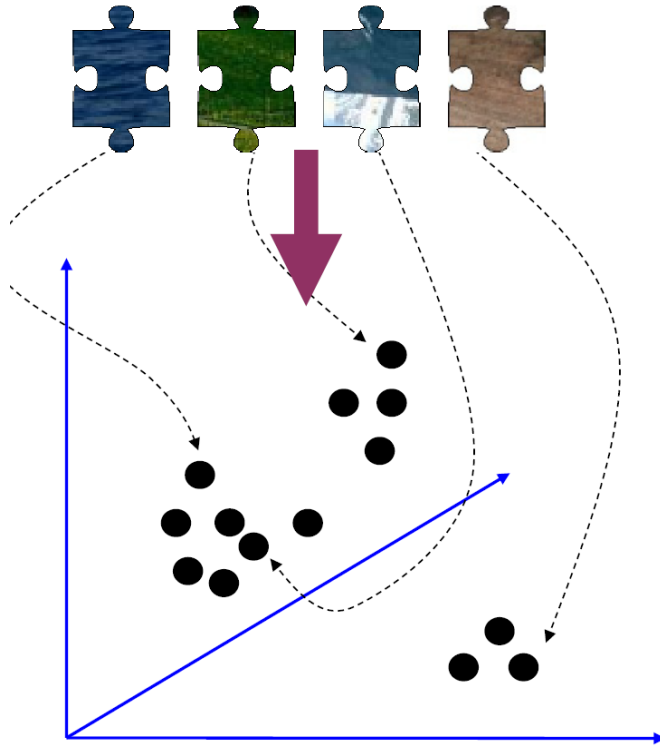
SURF

LBP



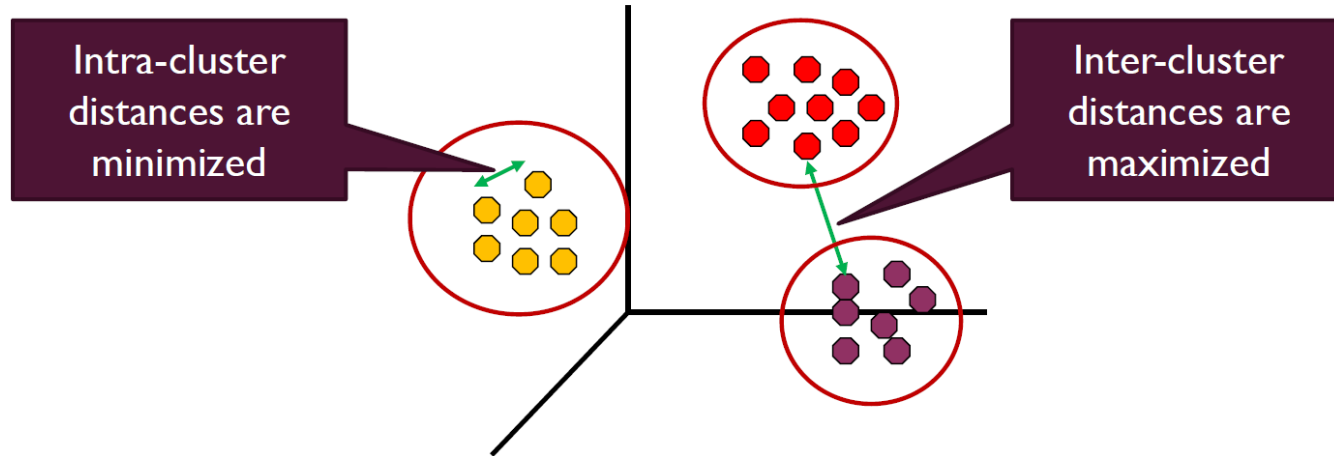
Feature Clustering

CLUSTERING FOR BOW

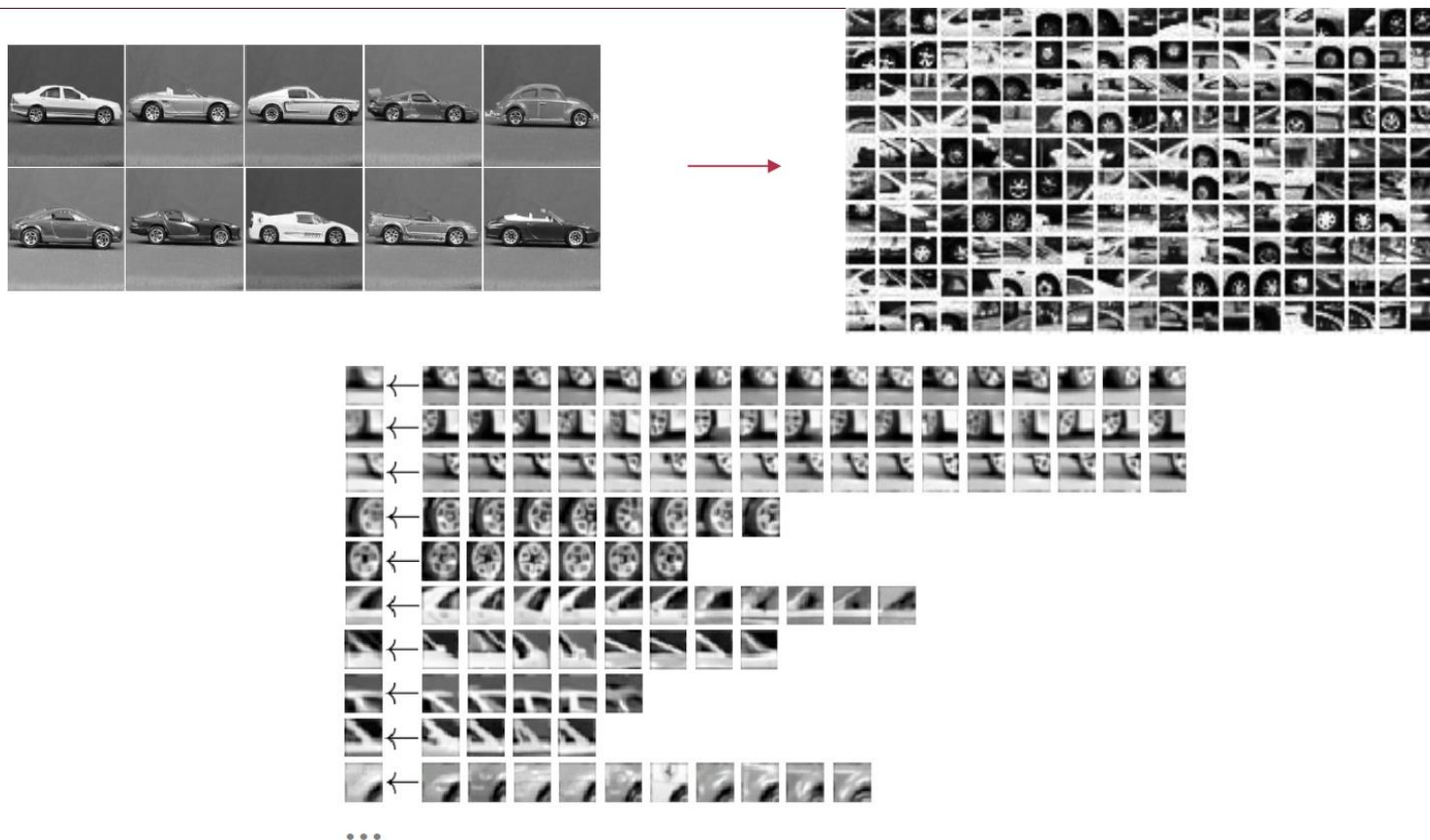


Feature Clustering

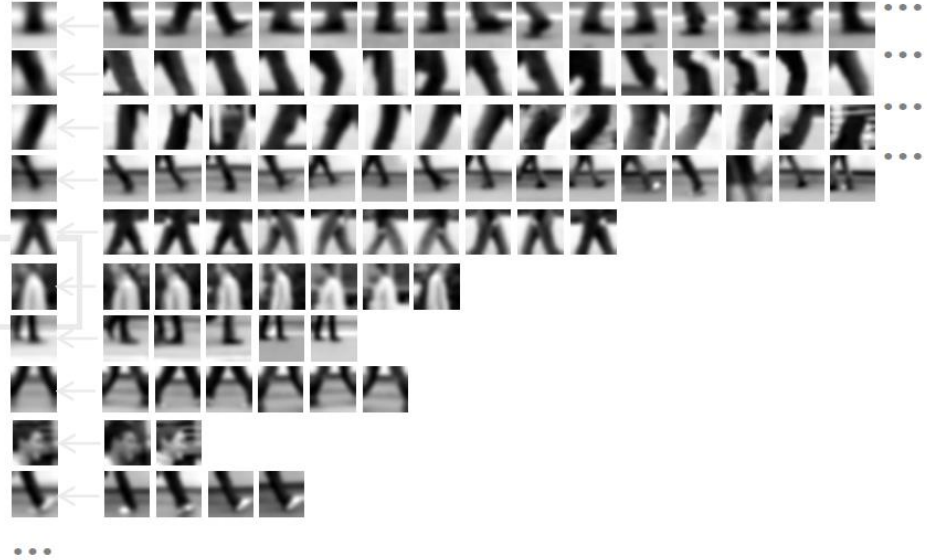
Find clusters of similar descriptors



Codebook Example



Codebook Example

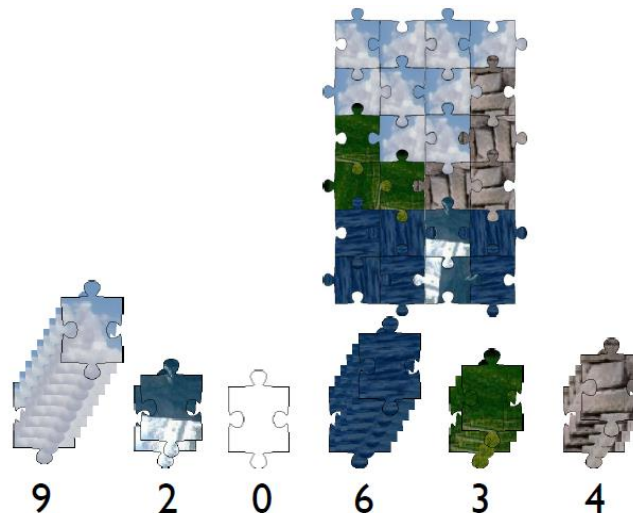


Appearance codebook

Histogram Creation

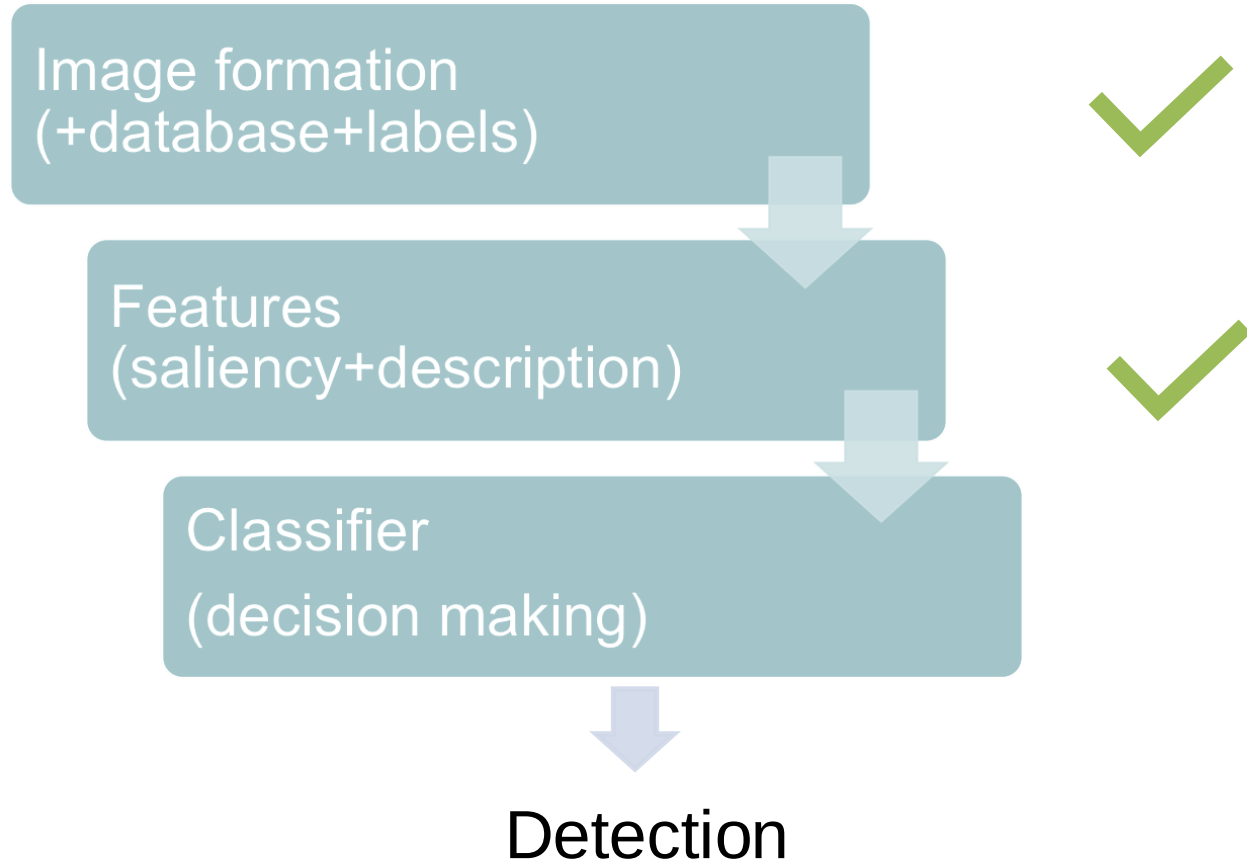
Simplest form:

- Assign each descriptor to nearest prototype
- Count how often each prototype appears

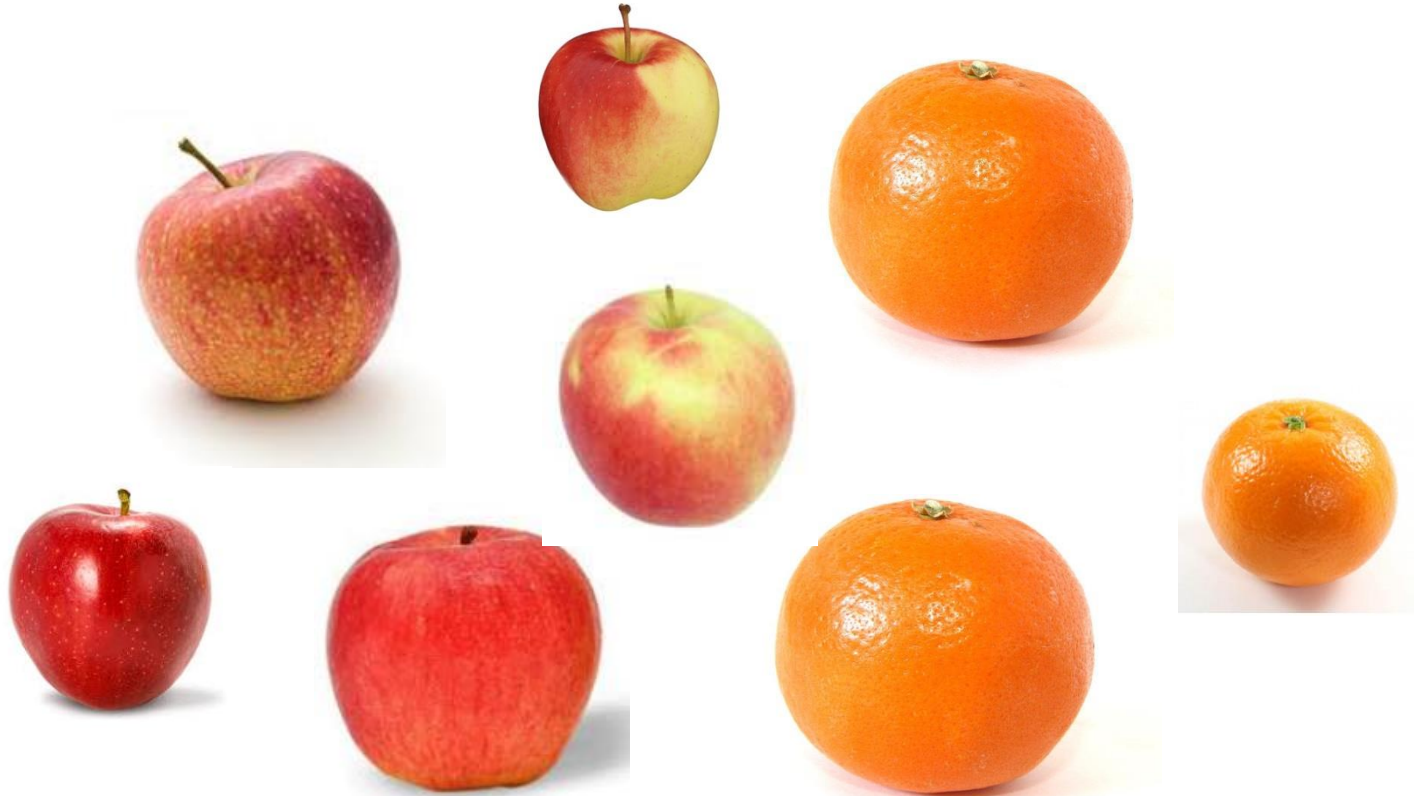


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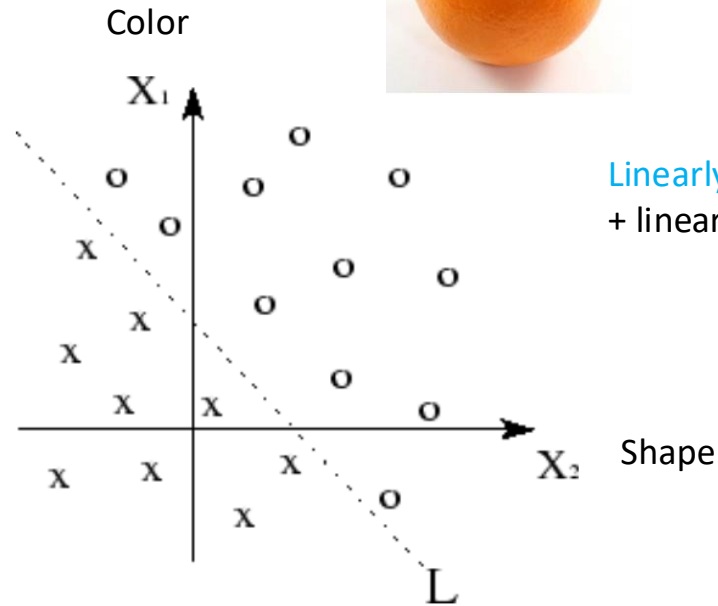
Classification Steps



Apple - Orange Classification

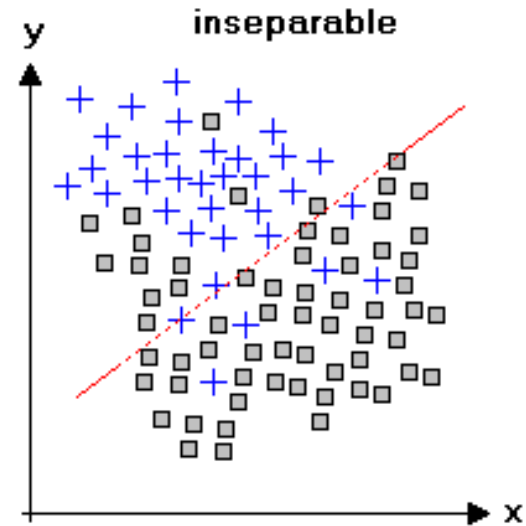
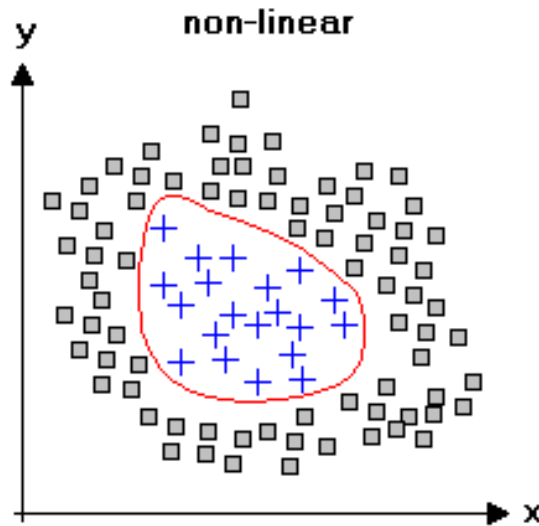
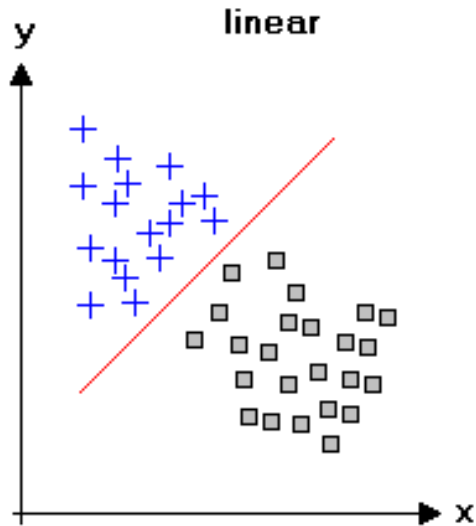


Feature Spaces - Separability

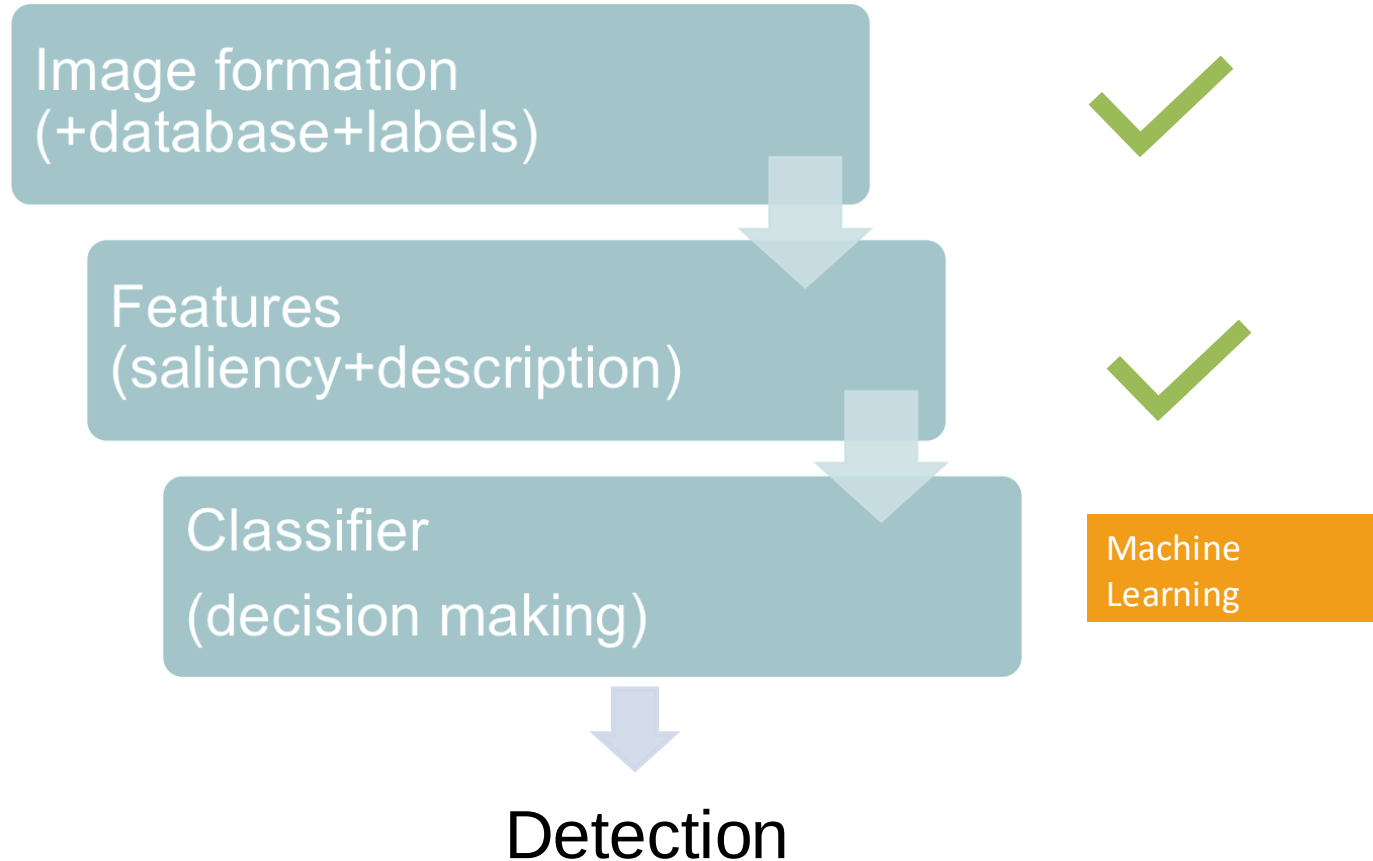


Linearly separable data
+ linear classifier = good.

Feature Spaces - Separability



Classification Steps



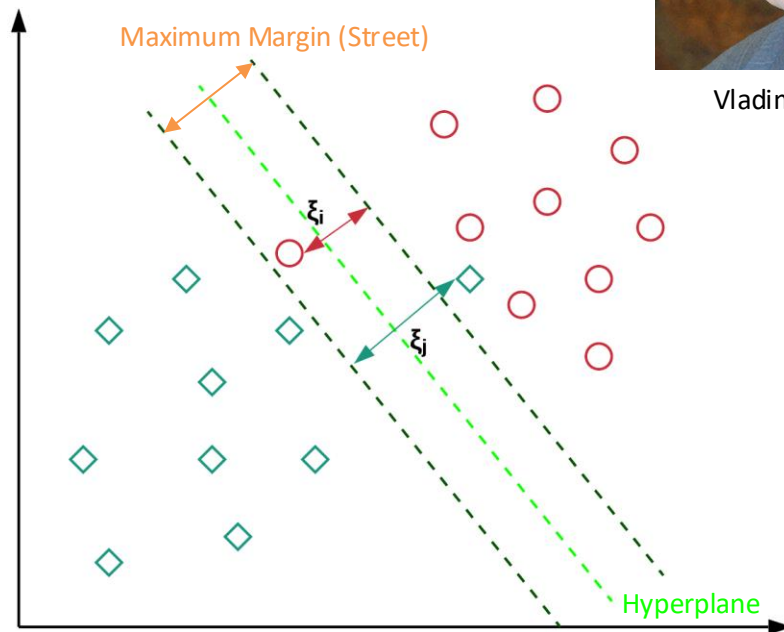
Support Vector Machines (SVMs)

- Goal: Find separating hyperplane (1D: point, 2D: line, 3D: plane, etc.)
- Maximizes margin (street) between classes
- Support vectors are data points which are closest to the decision boundary
- Only support vectors define the decision boundary (no other data points)
- Support vectors are found via optimization if linear separation exists (**Hard Margin SVM**)
- **Soft Margin SVMs** allow for outliers by additionally minimizing slack variables ξ_i that shift separation boundary
- Extension to non-linear decision boundaries via kernel trick (i.e. apply a non-linear transformation before SVM fitting).

SVMs work very well and are quite robust to noisy data!



Vladimir Vapnik



[Good explanation for kernel trick: <https://youtu.be/O7yT0-5VII>]

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- **Image Classification**
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- Accuracy:

$$\text{Accuracy} = \frac{\text{TN} + \text{TP}}{\text{TN} + \text{FP} + \text{TP} + \text{FN}}$$

- Precision and Recall:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

- F1 score: geometric mean of precision and recall

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Precision and Recall

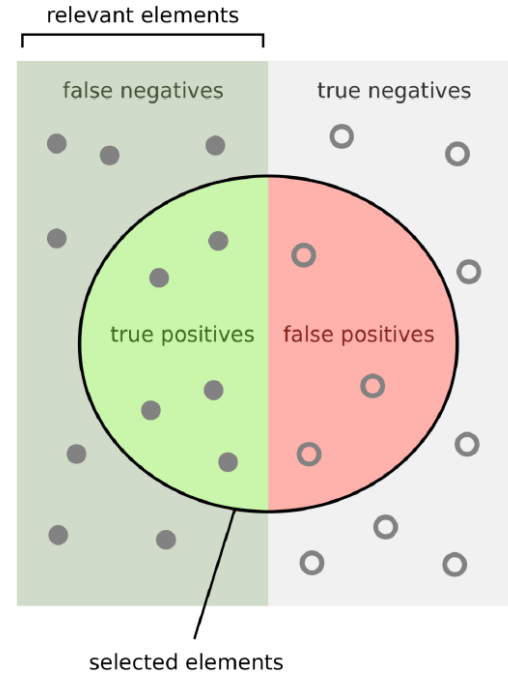
How many selected items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$


How many relevant items are selected?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$


Combine P+R into: F1-Measure



Accuracy

- Accuracy:

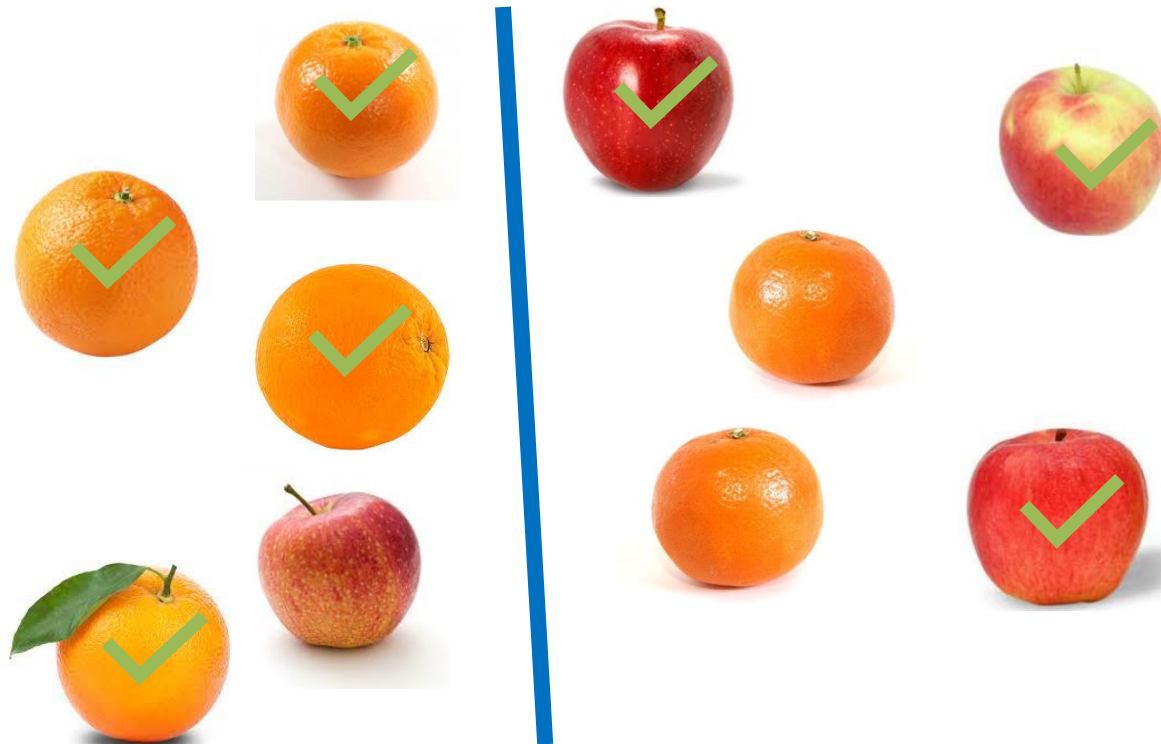
$$= \frac{\text{total correct}}{\text{total observations}}$$

$$= \frac{7}{10} = 70\%$$

- Problem with imbalanced classes, e.g.:

990 oranges & 10 apples

Model that predicts always oranges has 99 % accuracy.



Oranges

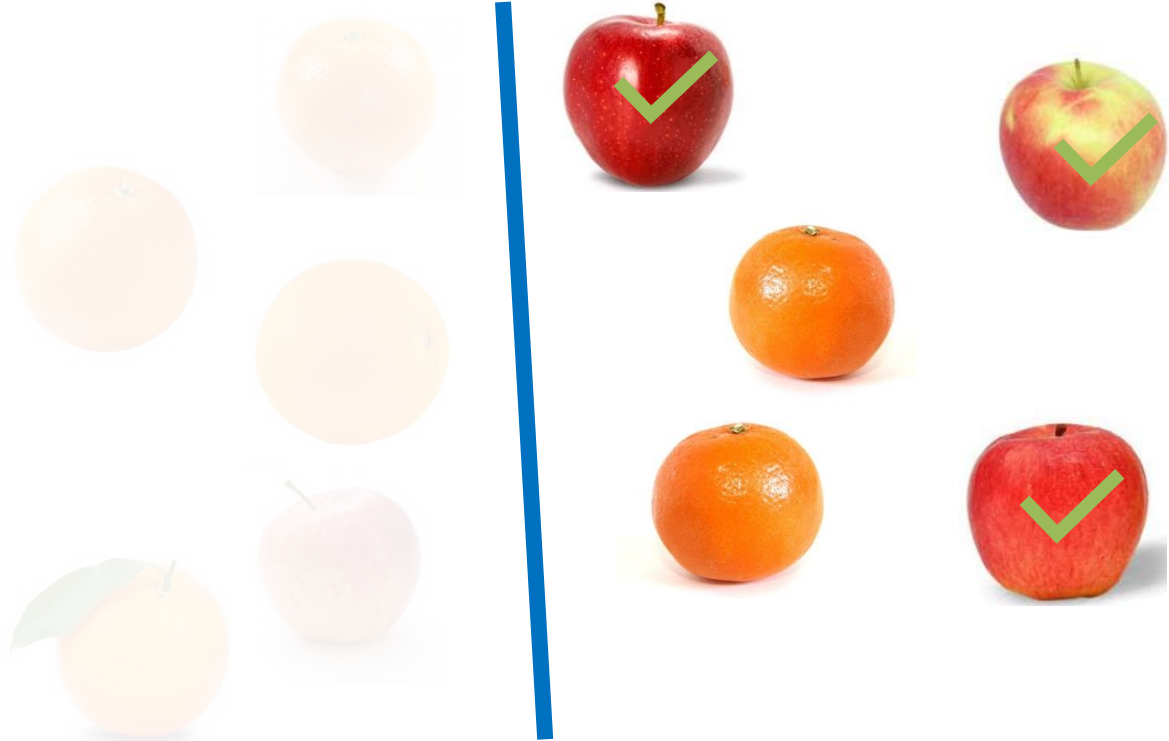
Apples

Precision

- Precision for apples

$$= \frac{\text{total apples correct}}{\text{total apple side observations}}$$

$$= \frac{3}{5} = 60\%$$



Oranges

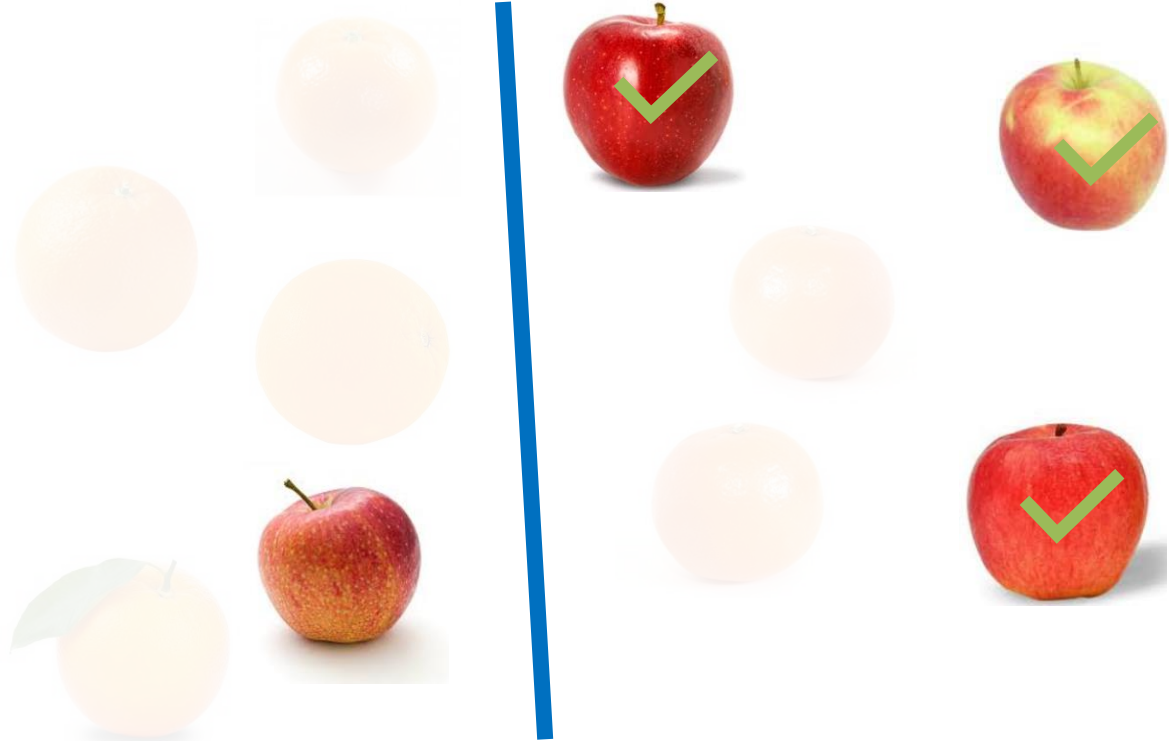
Apples

Recall

- Recall for apples

$$= \frac{\text{total apples correct}}{\text{all actual apples}}$$

$$= \frac{3}{4} = 75\%$$



Oranges

Apples

Change Decision Boundary

- Precision for apples

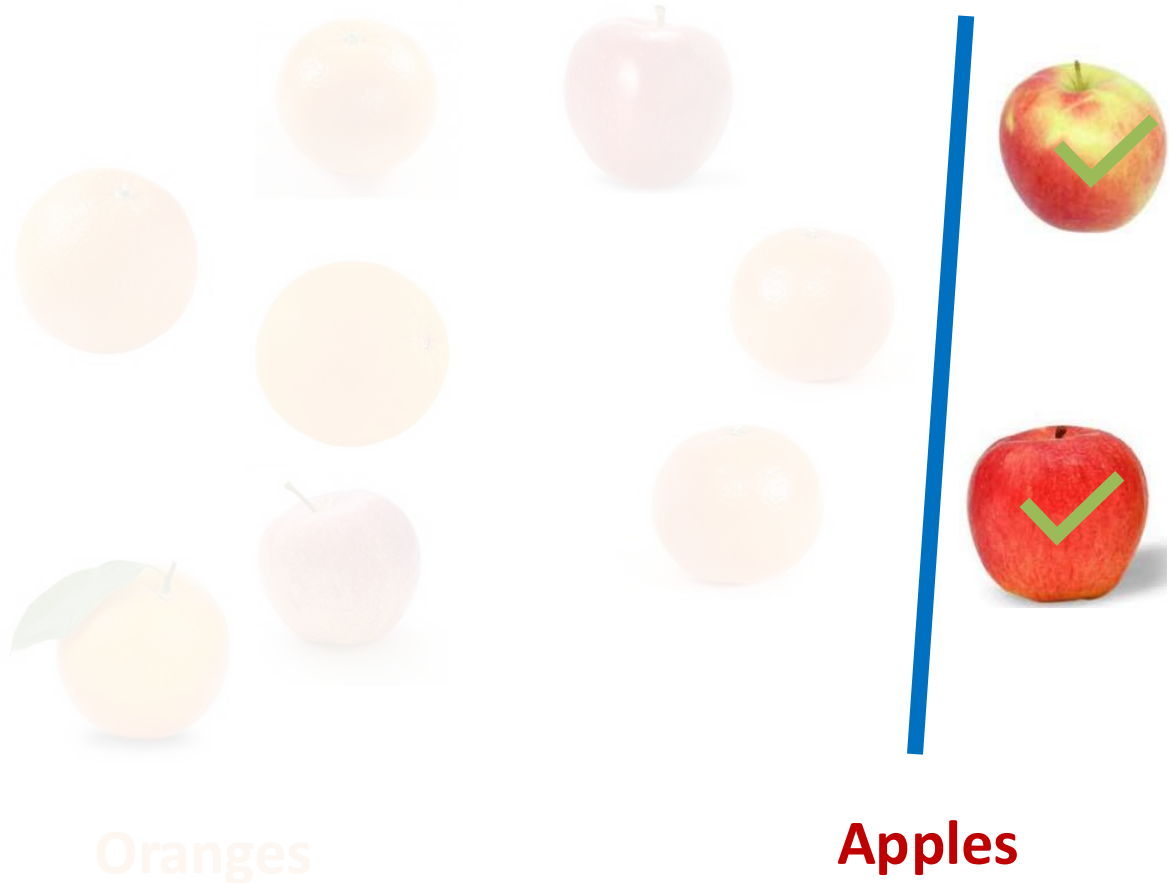
$$= \frac{\text{total apples correct}}{\text{total apple side observations}}$$

$$= \frac{2}{2} = 100\%$$

- Recall for apples

$$= \frac{\text{total apples correct}}{\text{all actual apples}}$$

$$= \frac{2}{4} = 50\%$$



Change Decision Boundary

- Precision for apples

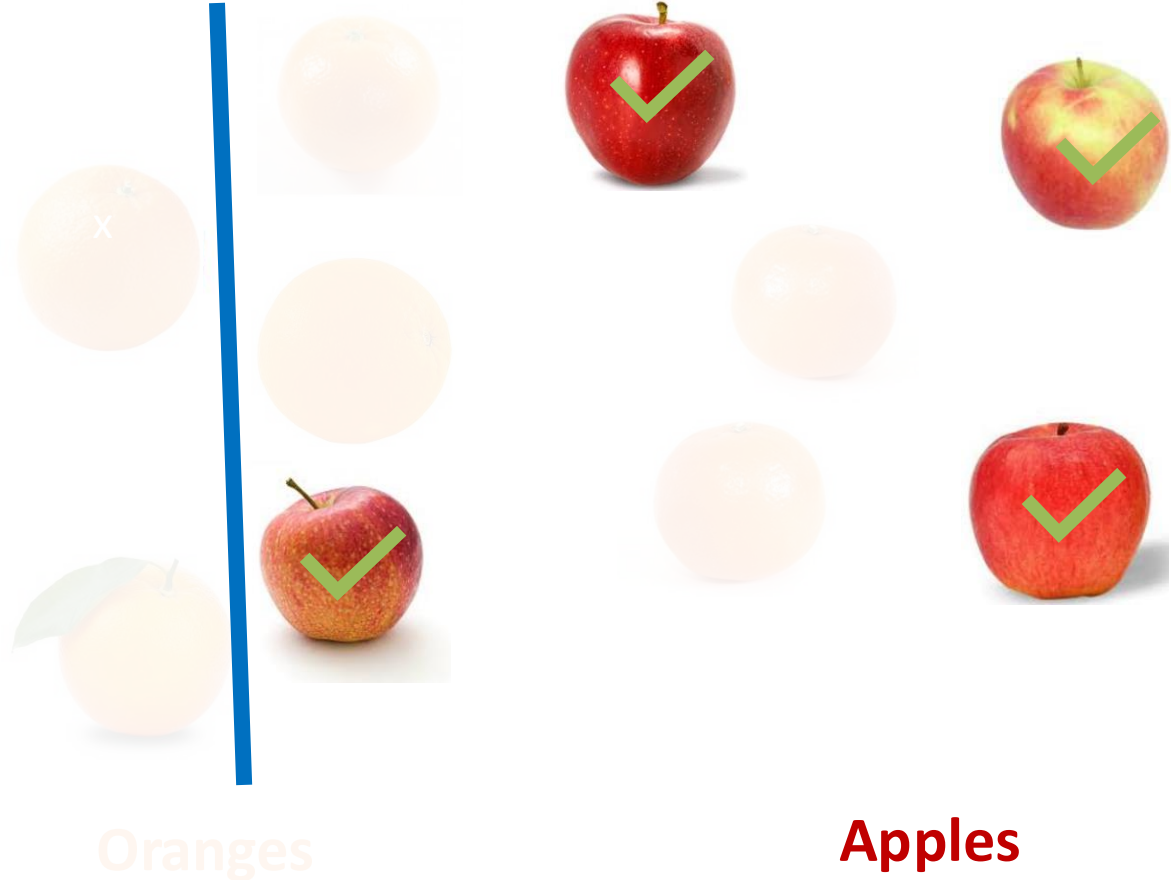
$$= \frac{\text{total apples correct}}{\text{total apple side observations}}$$

$$= \frac{4}{8} = 50\%$$

- Recall for apples

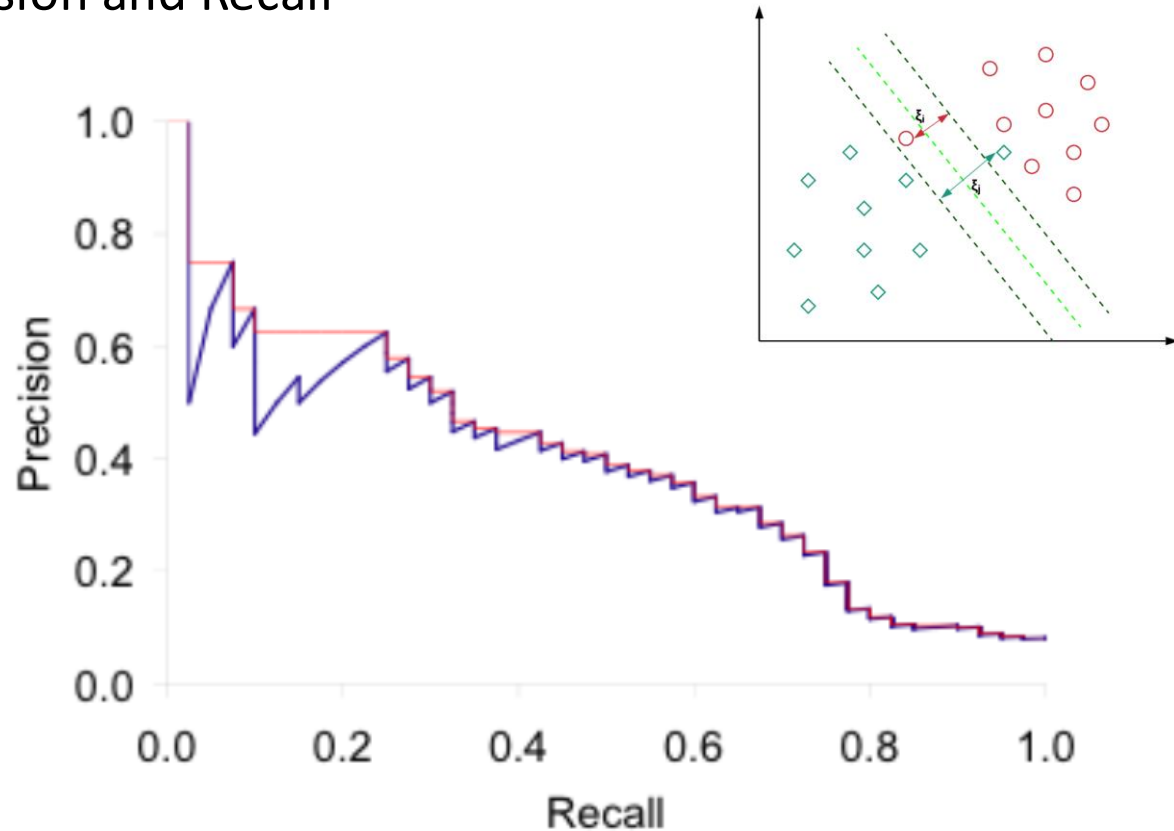
$$= \frac{\text{total apples correct}}{\text{all actual apples}}$$

$$= \frac{4}{4} = 100\%$$



Precision-Recall Curve

- Trade-off Precision and Recall

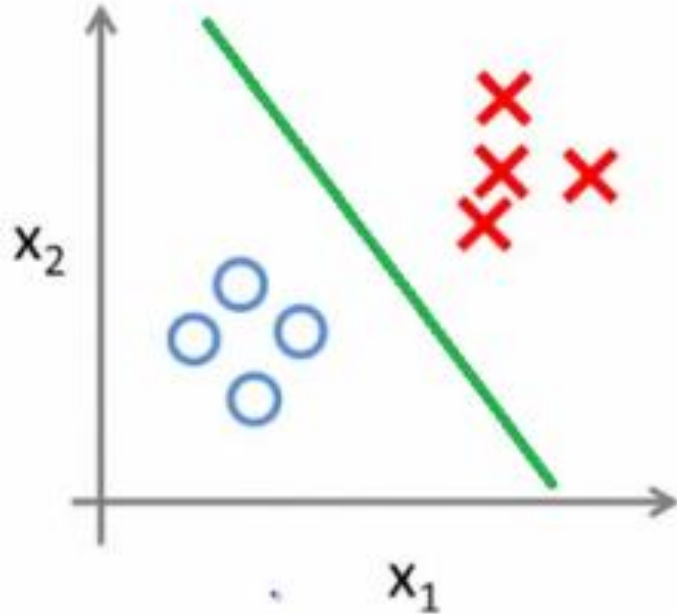


Today's Agenda

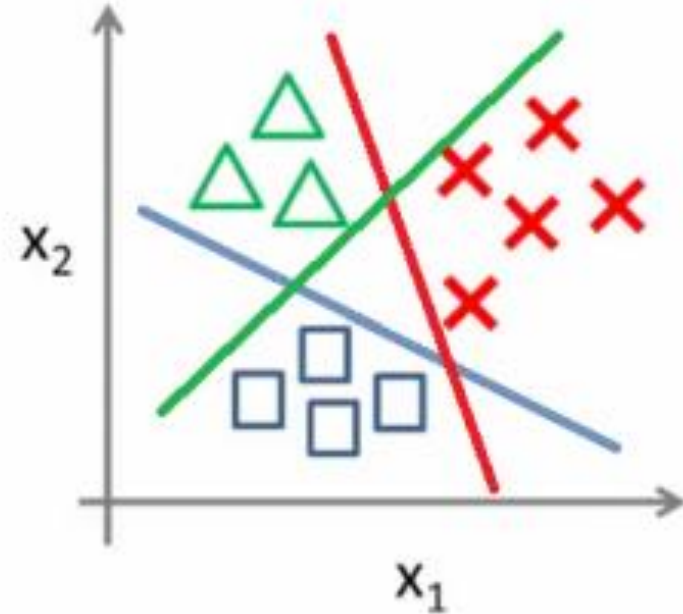
- **Image Classification**
 - Feature Representation
 - Bag of words
 - Feature space & Decision boundary
 - Classification Performance Evaluation
 - **Multi-class Classification**
 - Datasets

Multi-class Classification

Binary classification:



Multi-class classification:



- Simple multi-class extension: series of binary classification - one class against all other.

Today's Agenda



- **Image Classification**
 - Feature Representation
 - Bag of words
 - Feature space & Decision boundary
 - Classification Performance Evaluation
 - Multi-class Classification
 - **Datasets**

Datasets



22K categories and **15M** images

- Animals
 - Bird
 - Fish
 - Mammal
 - Invertebrate
- Plants
 - Tree
 - Flower
- Food
- Materials
- Structures
 - Artifact
 - Tools
 - Appliances
 - Structures
- Person
- Scenes
 - Indoor
 - Geological Formations
- Sport Activity

www.image-net.org

Deng et al. 2009,
Russakovsky et al. 2015



Slide credit: Fei-Fei Li

What is WordNet?



Original paper
by
**[George
Miller, et al
1990]** cited
over 5,000
times

Organizes over
150,000 words
into 117,000
categories
called *synsets*.

Establishes
ontological and
lexical
relationships in
NLP and related
tasks.

Individually Illustrated WordNet Nodes



jacket: a short coat



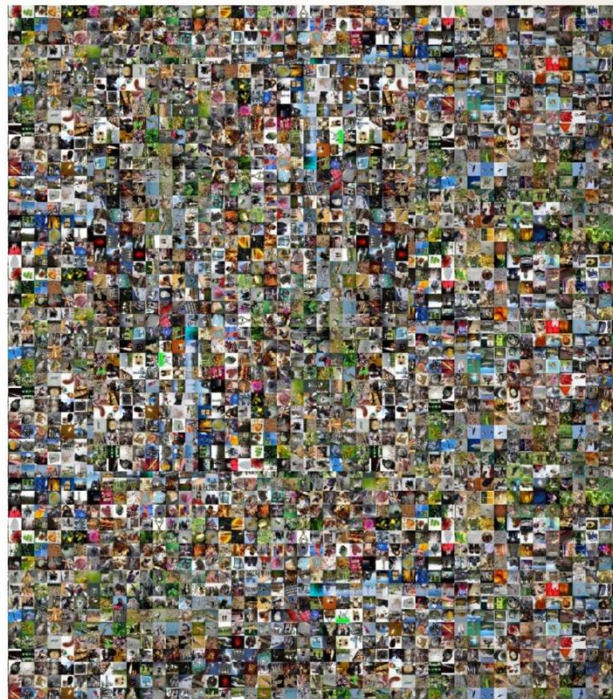
German shepherd:
breed of large shepherd
dogs used in police work
and as a guide for the
blind.



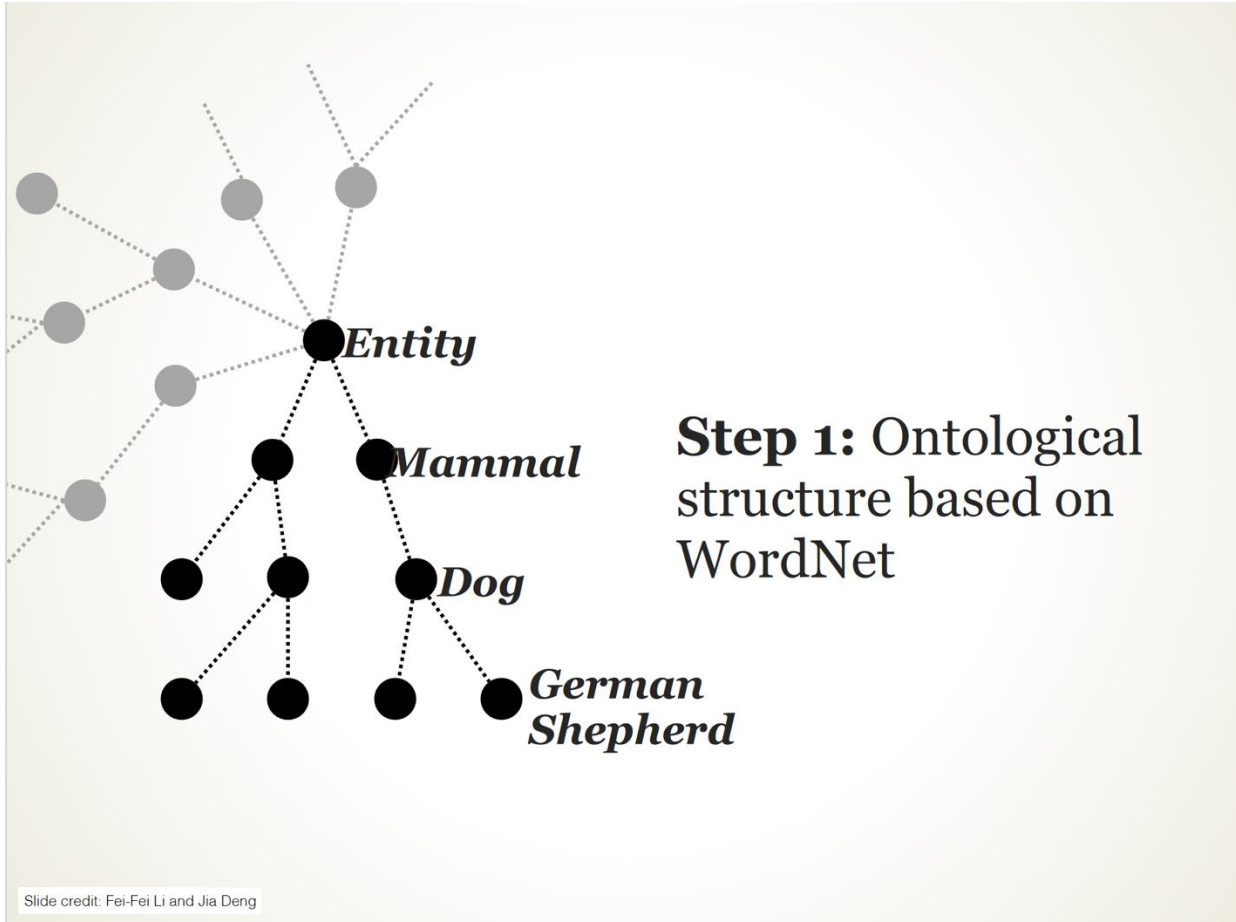
microwave: kitchen
appliance that cooks food
by passing an
electromagnetic wave
through it.



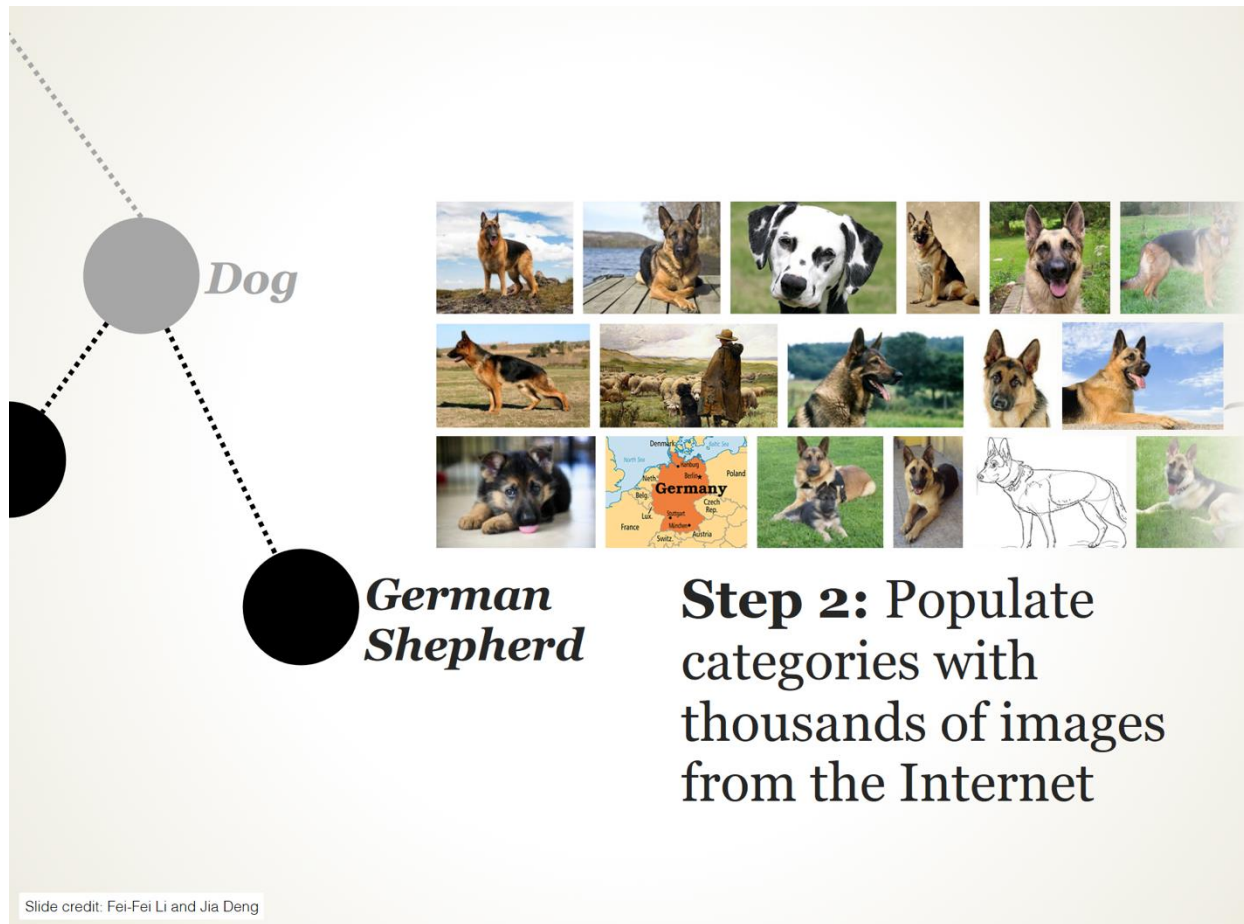
mountain: a land
mass that projects well
above its surroundings;
higher than a hill.



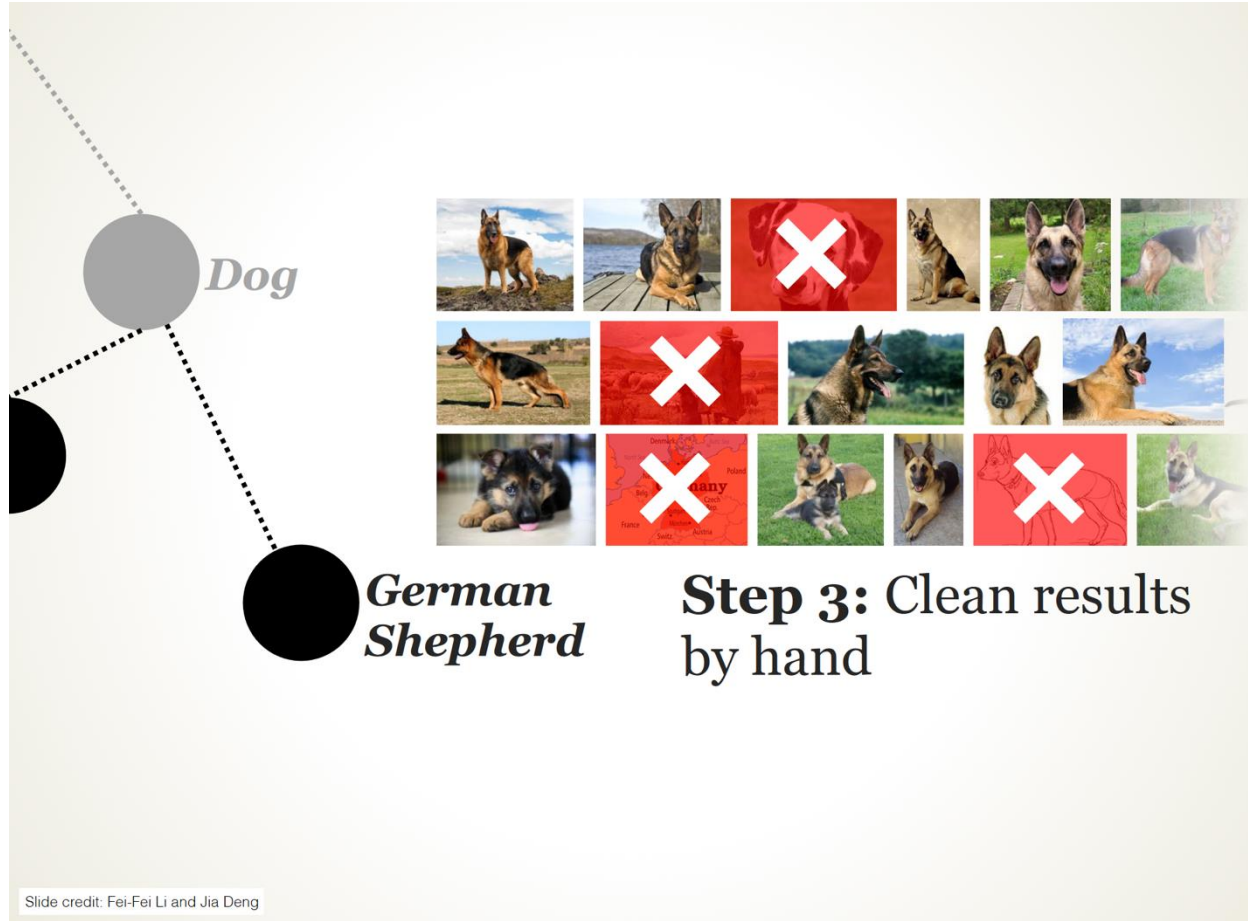
Datasets



Datasets



Datasets



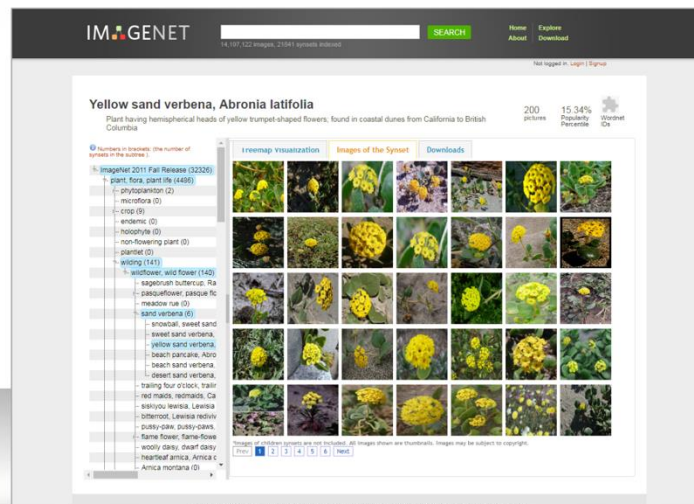
3rd Attempt: Crowdsourcing

ImageNet
PhD
Students

Crowdsourced
Labor



The Result: IMAGENET Goes Live in 2009



ImageNet Targeted Scale

SUN, 131K

[Xiao et al. '10]

LabelMe, 37K

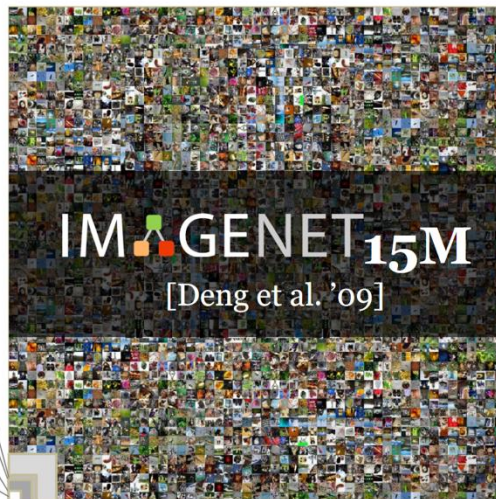
[Russell et al. '07]

PASCAL VOC, 30K

[Everingham et al. '06-'12]

Caltech101, 9K

[Fei-Fei, Fergus, Perona, '03]



More Resources

Papers with code <https://paperswithcode.com/datasets?task=image-classification>

Best match

Filter by Modality

Images

139

Texts

13

Medical

11

Biomedical

4

Graphs

4

Hyperspectral images

3

Videos

3

Audio

2

Filter by Task (clear)

Image Classification

292

Question Answering

221

Semantic Segmentation

210

Object Detection

180

Speech Recognition

130

Language Modelling

102

Text Classification

177 dataset results for Image Classification



CIFAR-10

The CIFAR-10 dataset (Canadian Institute for Advanced Research, 10 classes) is a subset of the Tiny Images dataset and consists of 60000 32x32 color images. The images are labelled with...

9,361 PAPERS • 58 BENCHMARKS



ImageNet

The ImageNet dataset contains 14,197,122 annotated images according to the WordNet hierarchy. Since 2010 the dataset is used in the ImageNet Large Scale Visual Recognition Challenge...

8,932 PAPERS • 86 BENCHMARKS



MNIST

The MNIST database (Modified National Institute of Standards and Technology database) is a large collection of handwritten digits. It has a training set of 60,000 examples, and a test set of...

5,506 PAPERS • 43 BENCHMARKS



CIFAR-100

The CIFAR-100 dataset (Canadian Institute for Advanced Research, 100 classes) is a subset of the Tiny Images dataset and consists of 60000 32x32 color images. The 100 classes in the...

4,612 PAPERS • 34 BENCHMARKS



SVHN (Street View House Numbers)

Street View House Numbers (SVHN) is a digit classification benchmark dataset that contains 600,000 32x32 RGB images of printed digits (from 0 to 9) cropped from pictures of house num...

2,254 PAPERS • 10 BENCHMARKS



CelebA (CelebFaces Attributes Dataset)

CelebFaces Attributes dataset contains 202,599 face images of the size 178x218 from 10,177 celebrities, each annotated with 40 binary labels indicating facial attributes like hair color, gen...

2,186 PAPERS • 15 BENCHMARKS



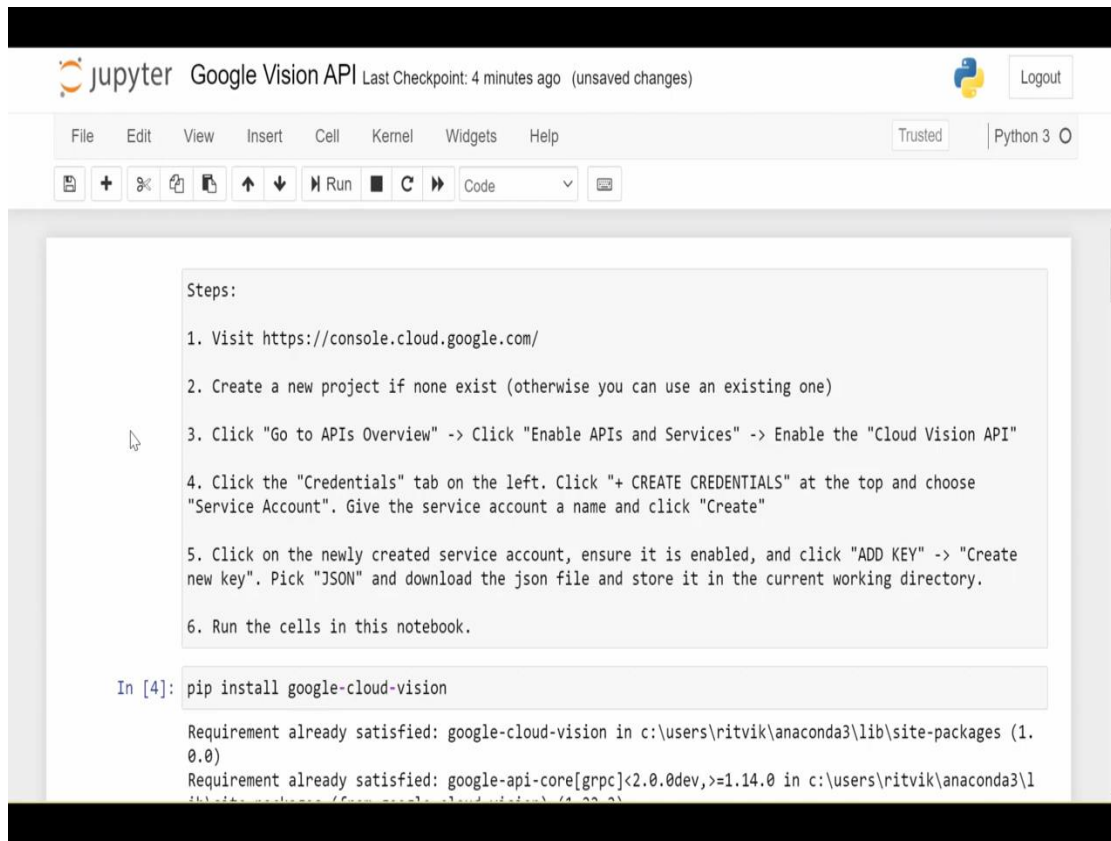
Fashion-MNIST

Fashion-MNIST is a dataset comprising of 28x28 grayscale images of 70,000 fashion products from 10 categories, with 7,000 images per category. The training set has 60,000 images and th...

1,925 PAPERS • 12 BENCHMARKS

More Resources

Google Vision API <https://www.youtube.com/watch?v=9FMVZkxfrfA>



The screenshot shows a Jupyter Notebook interface with the title 'Google Vision API'. The top bar indicates 'Last Checkpoint: 4 minutes ago (unsaved changes)' and includes a 'Logout' button. The menu bar includes 'File', 'Edit', 'View', 'Insert', 'Cell', 'Kernel', 'Widgets', and 'Help'. The toolbar shows icons for saving, adding cells, and running code. The notebook content consists of a text cell with a list of steps and a code cell with a terminal output.

Steps:

1. Visit <https://console.cloud.google.com/>
2. Create a new project if none exist (otherwise you can use an existing one)
3. Click "Go to APIs Overview" -> Click "Enable APIs and Services" -> Enable the "Cloud Vision API"
4. Click the "Credentials" tab on the left. Click "+ CREATE CREDENTIALS" at the top and choose "Service Account". Give the service account a name and click "Create"
5. Click on the newly created service account, ensure it is enabled, and click "ADD KEY" -> "Create new key". Pick "JSON" and download the json file and store it in the current working directory.
6. Run the cells in this notebook.

In [4]: `pip install google-cloud-vision`

Requirement already satisfied: google-cloud-vision in c:\users\ritvik\anaconda3\lib\site-packages (1.0.0)
Requirement already satisfied: google-api-core[grpc]<2.0.0dev,>=1.14.0 in c:\users\ritvik\anaconda3\lib\site-packages (1.0.0)

Questions?

Disclaimer

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